

BFS Traversal

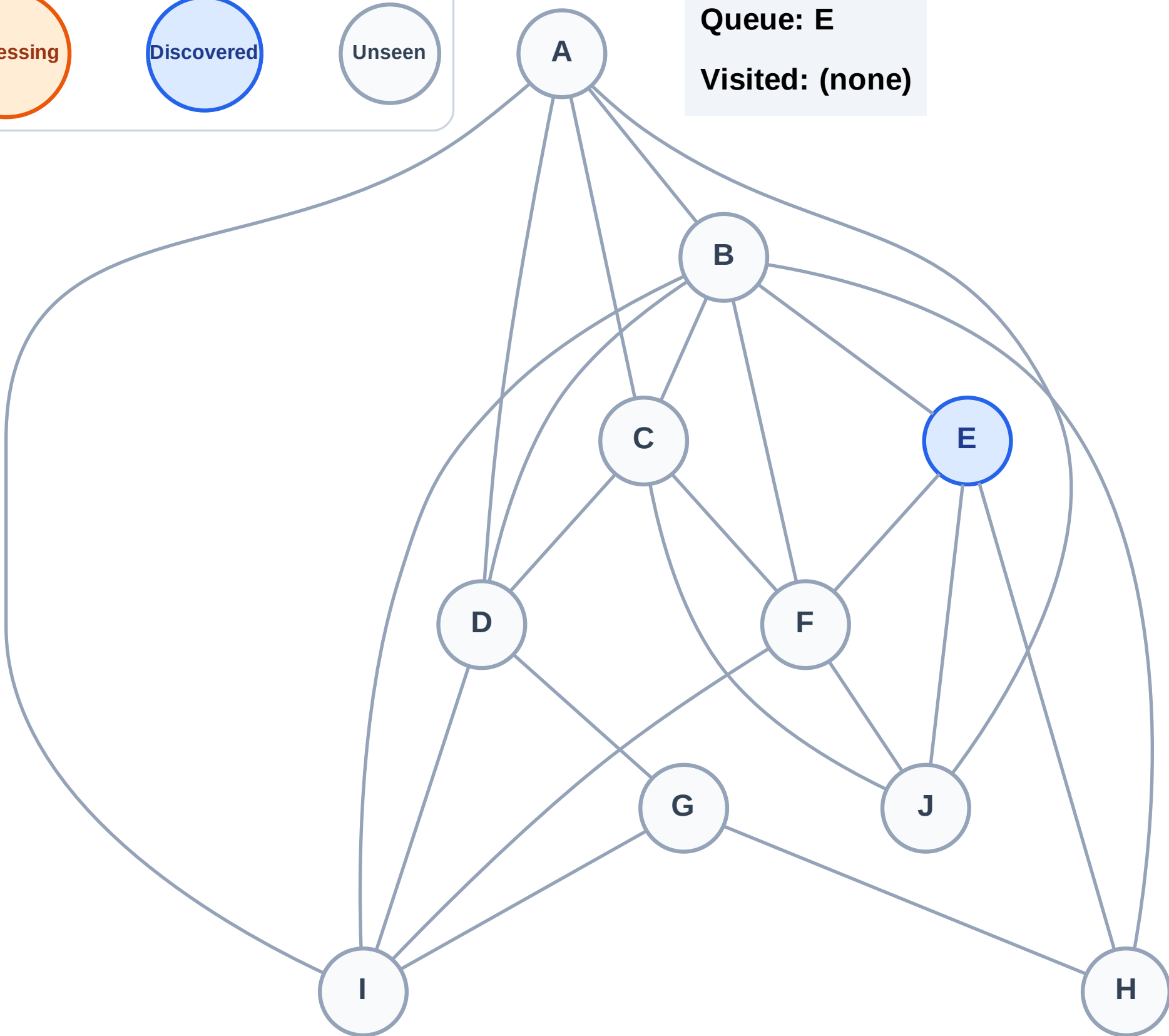
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

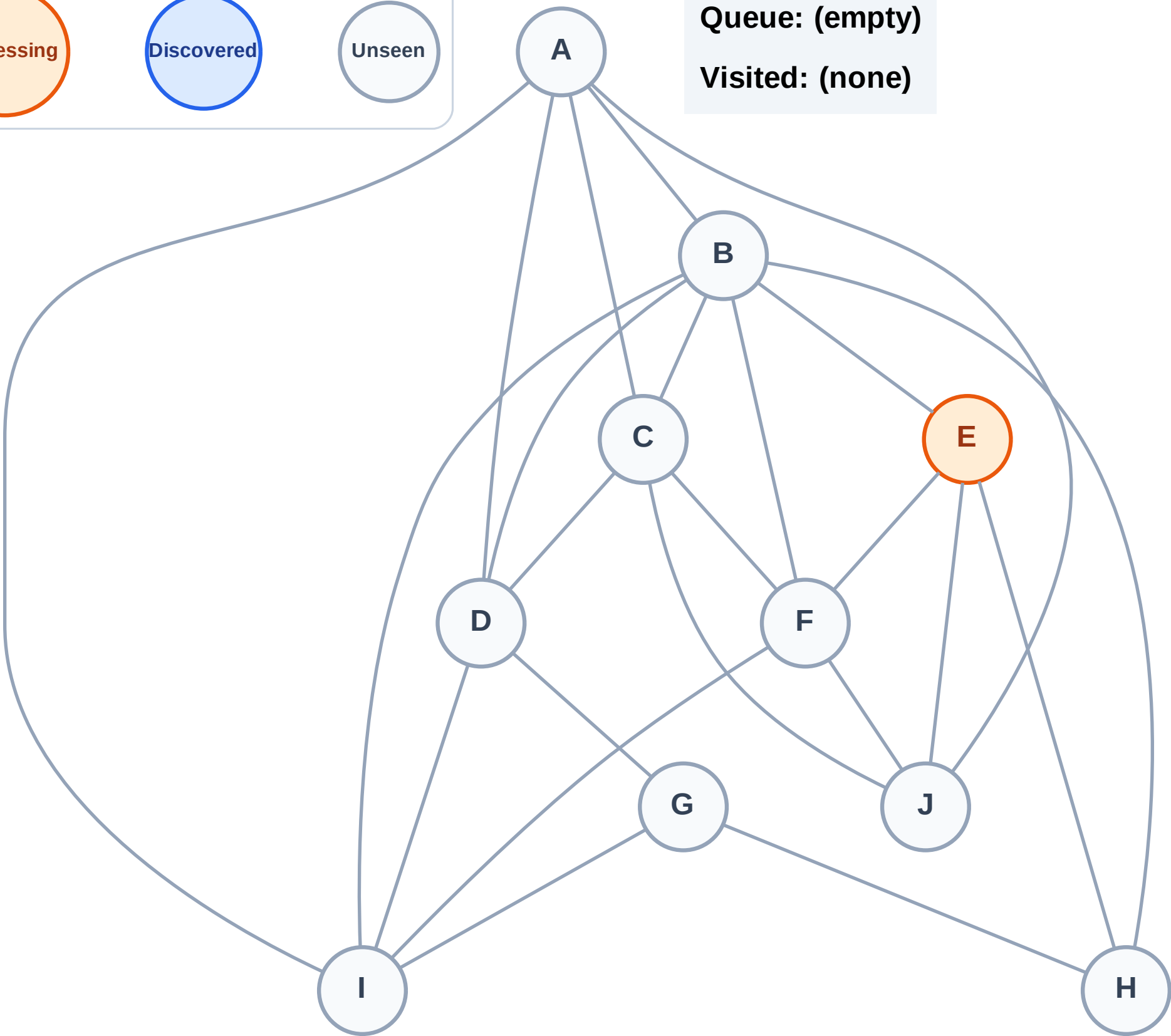
Complete

Processing

Discovered

Unseen

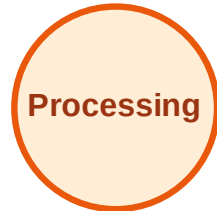
Queue: (empty)
Visited: (none)



BFS Traversal

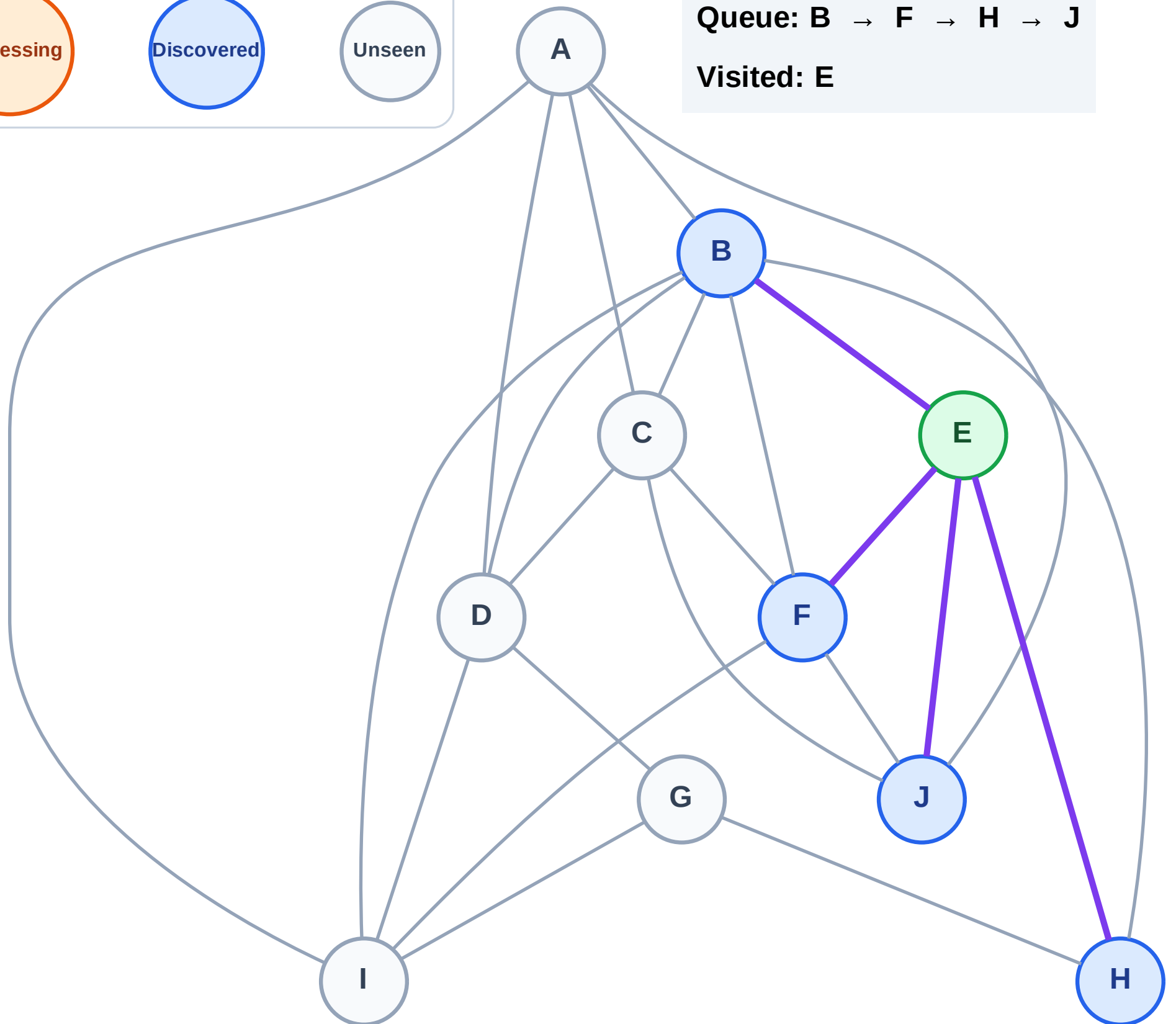
Step 3: Discover B, F, H, J from E

Legend



Queue: B → F → H → J

Visited: E



BFS Traversal

Step 4: Process node B

Complete

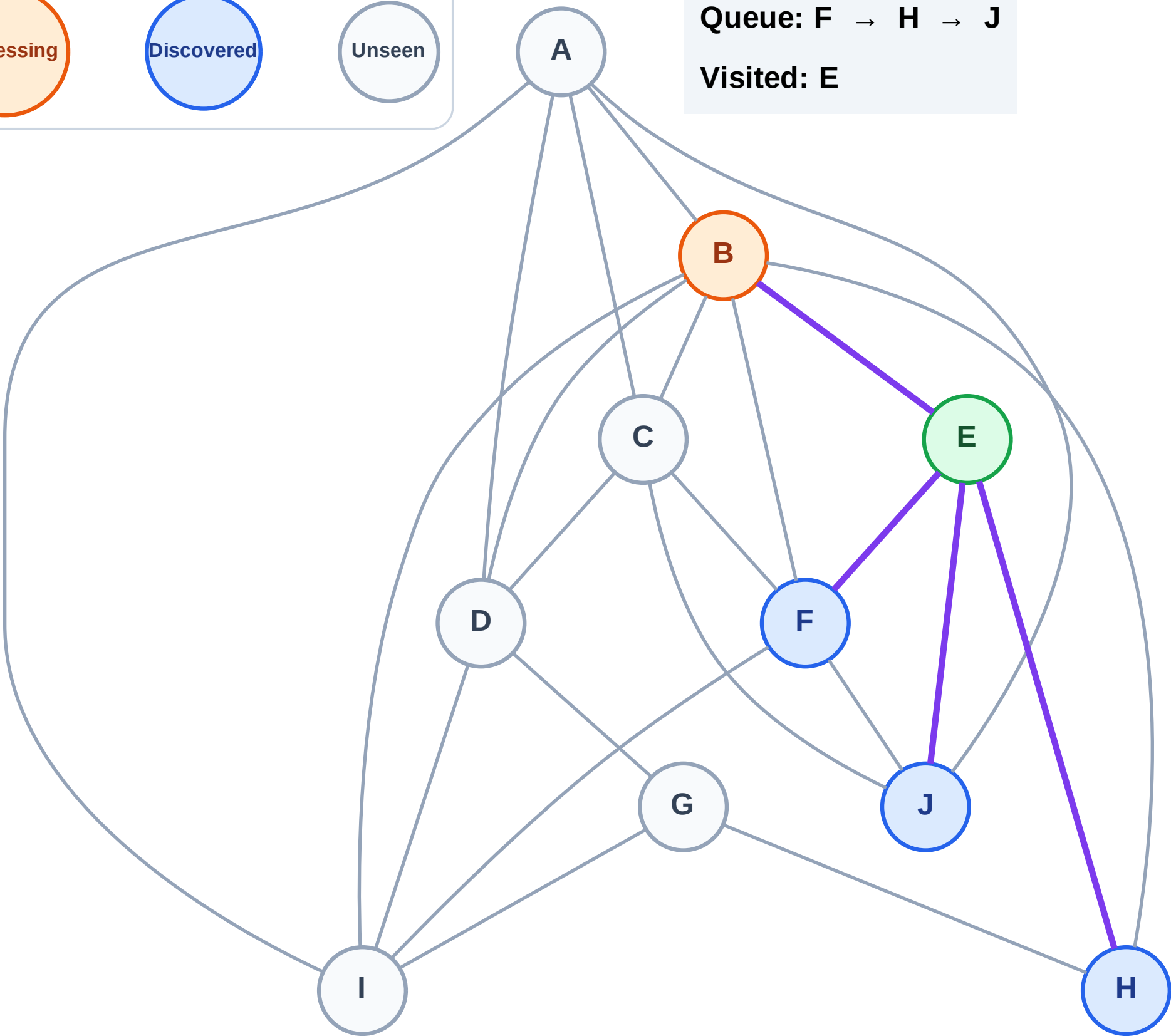
Processing

Discovered

Unseen

Queue: F → H → J

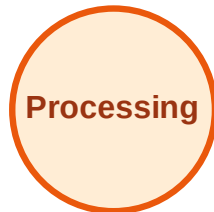
Visited: E



BFS Traversal

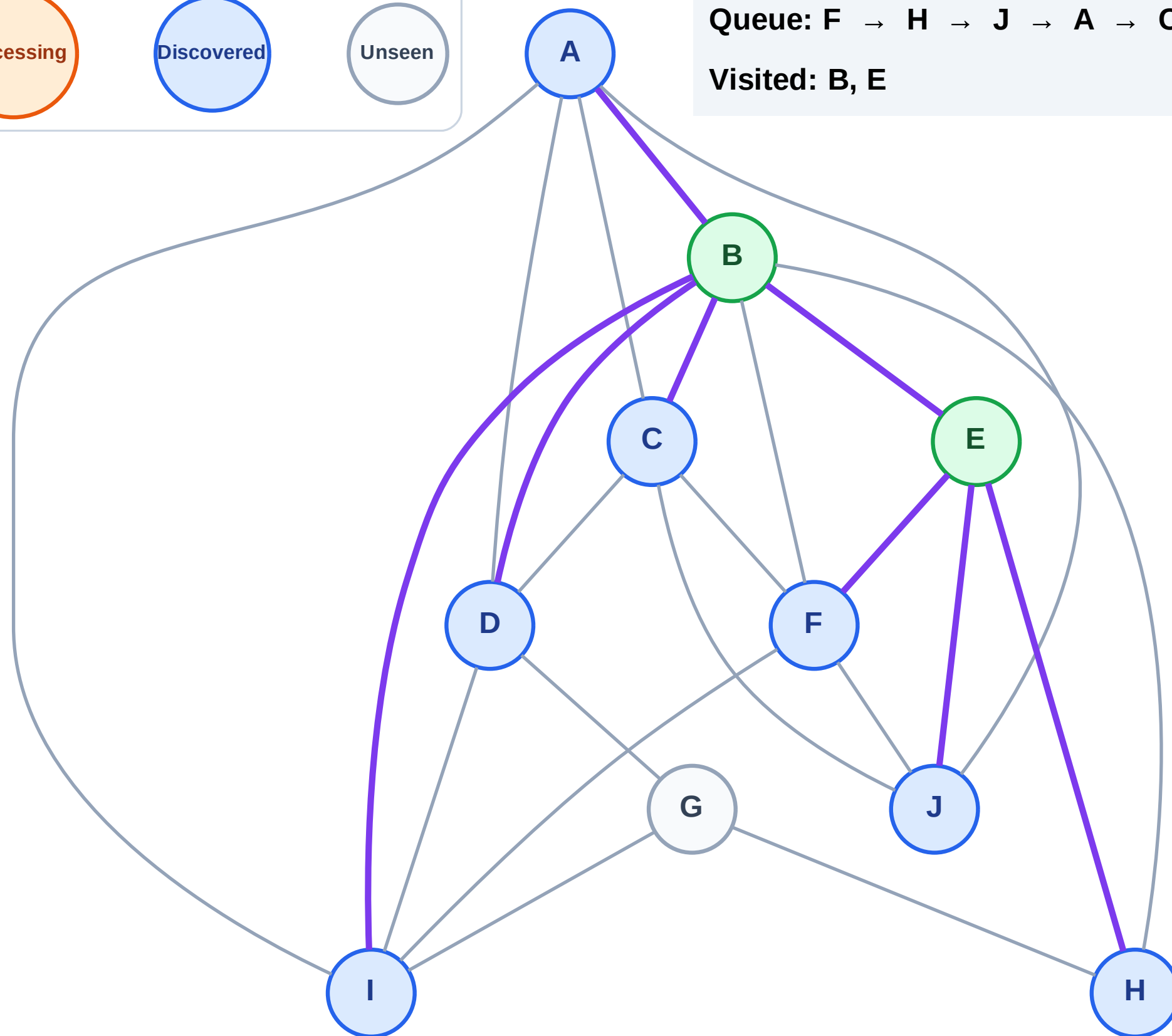
Step 5: Discover A, C, D, I from B

Legend



Queue: F → H → J → A → C → D → I

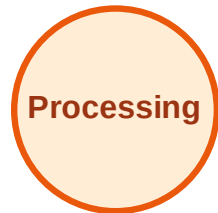
Visited: B, E



BFS Traversal

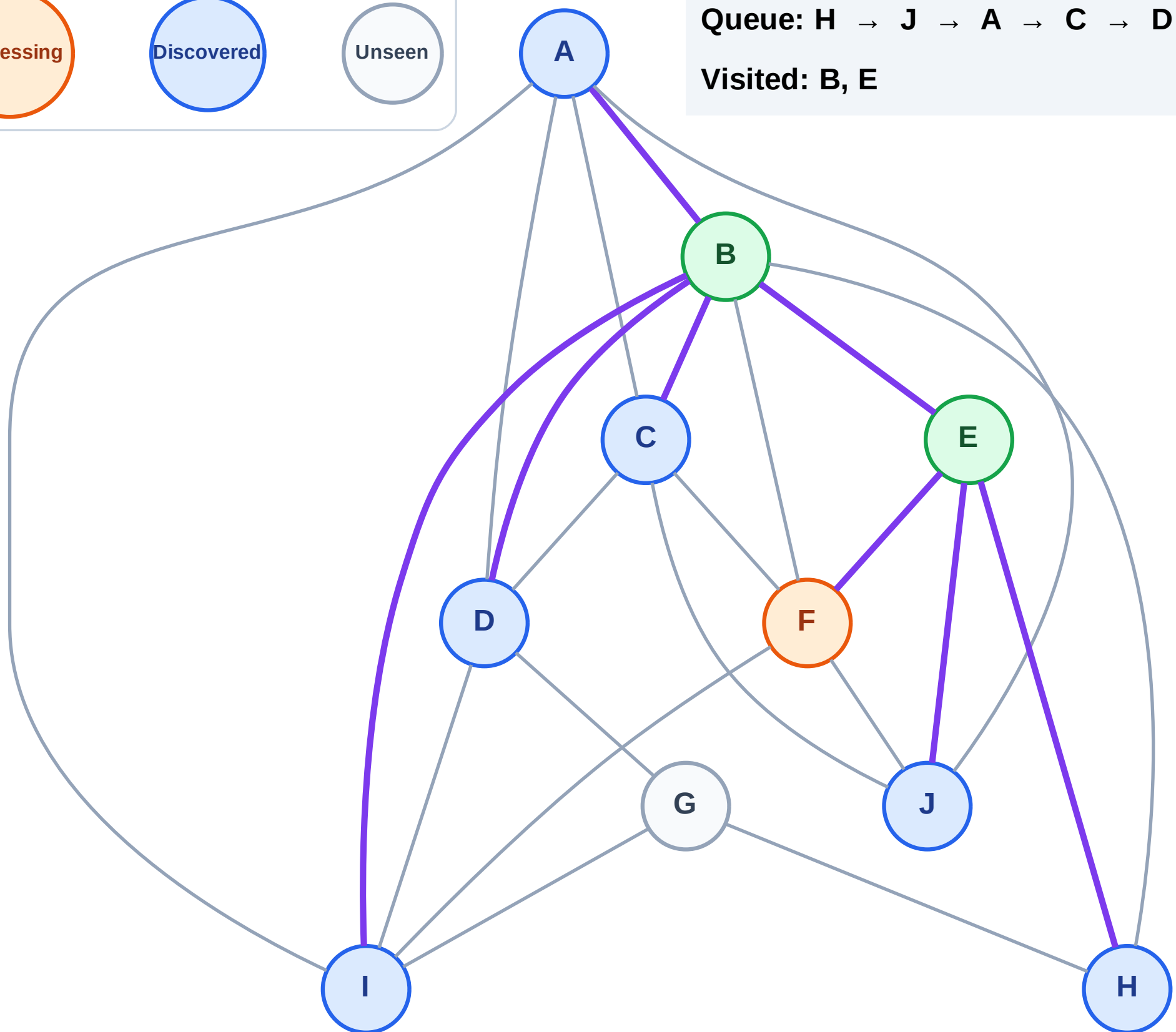
Step 6: Process node F

Legend



Queue: H → J → A → C → D → I

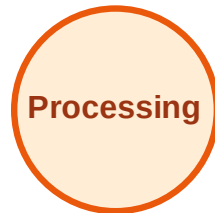
Visited: B, E



BFS Traversal

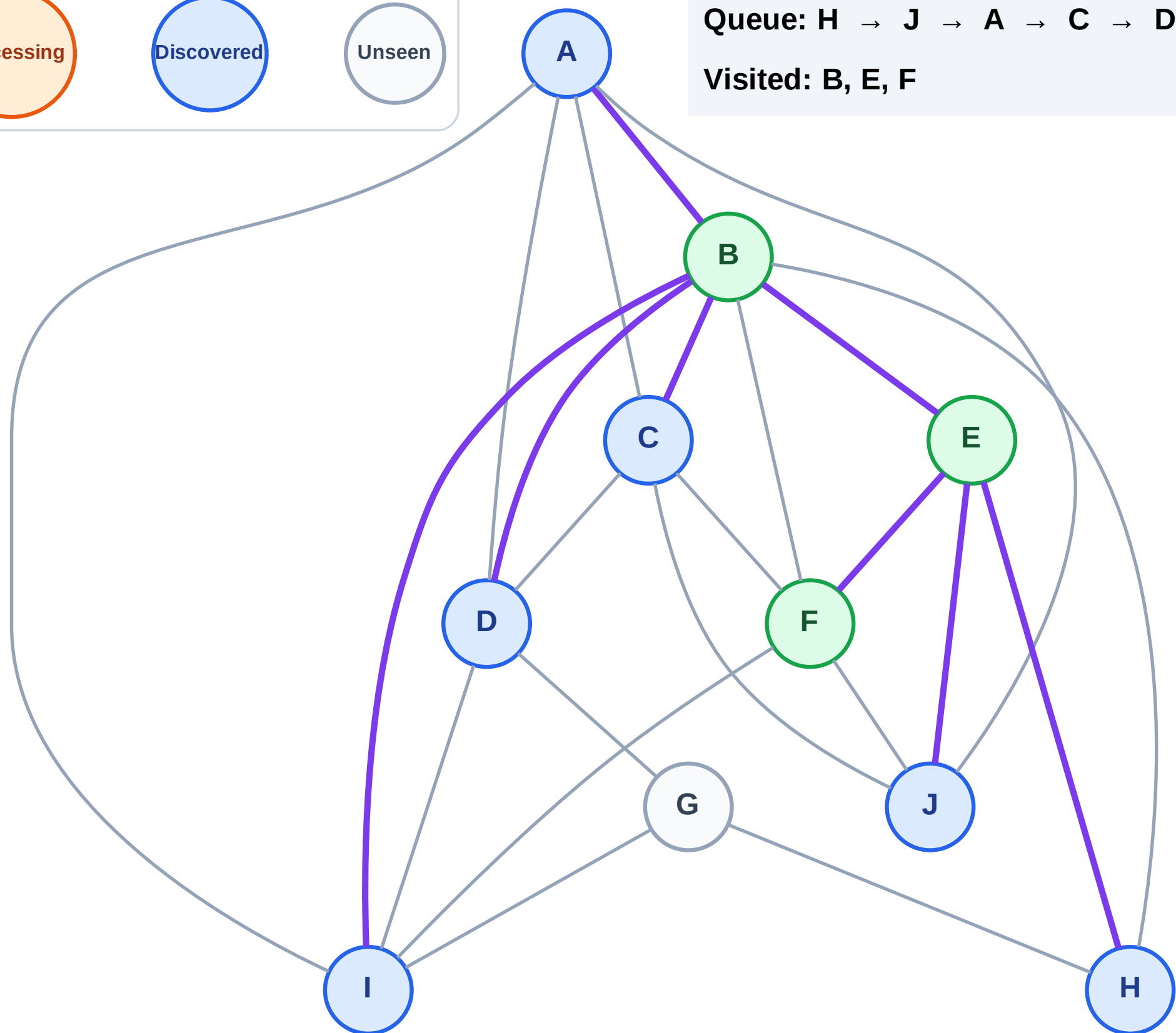
Step 7: No new neighbors from F

Legend



Queue: H → J → A → C → D → I

Visited: B, E, F



BFS Traversal

Step 8: Process node H

Legend

Complete

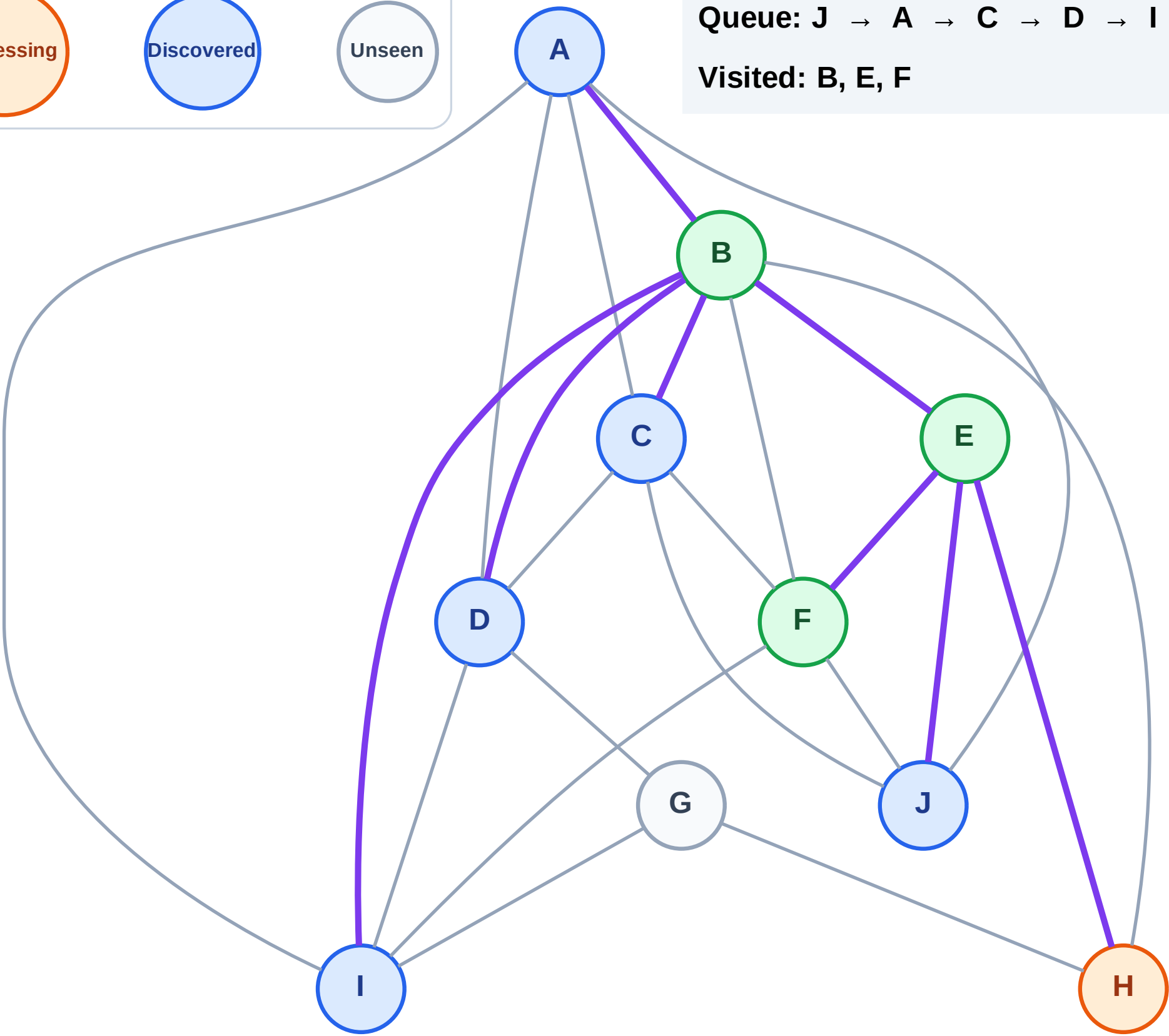
Processing

Discovered

Unseen

Queue: J → A → C → D → I

Visited: B, E, F



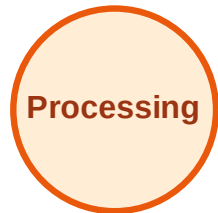
BFS Traversal

Step 9: Discover G from H

Legend



Complete



Processing



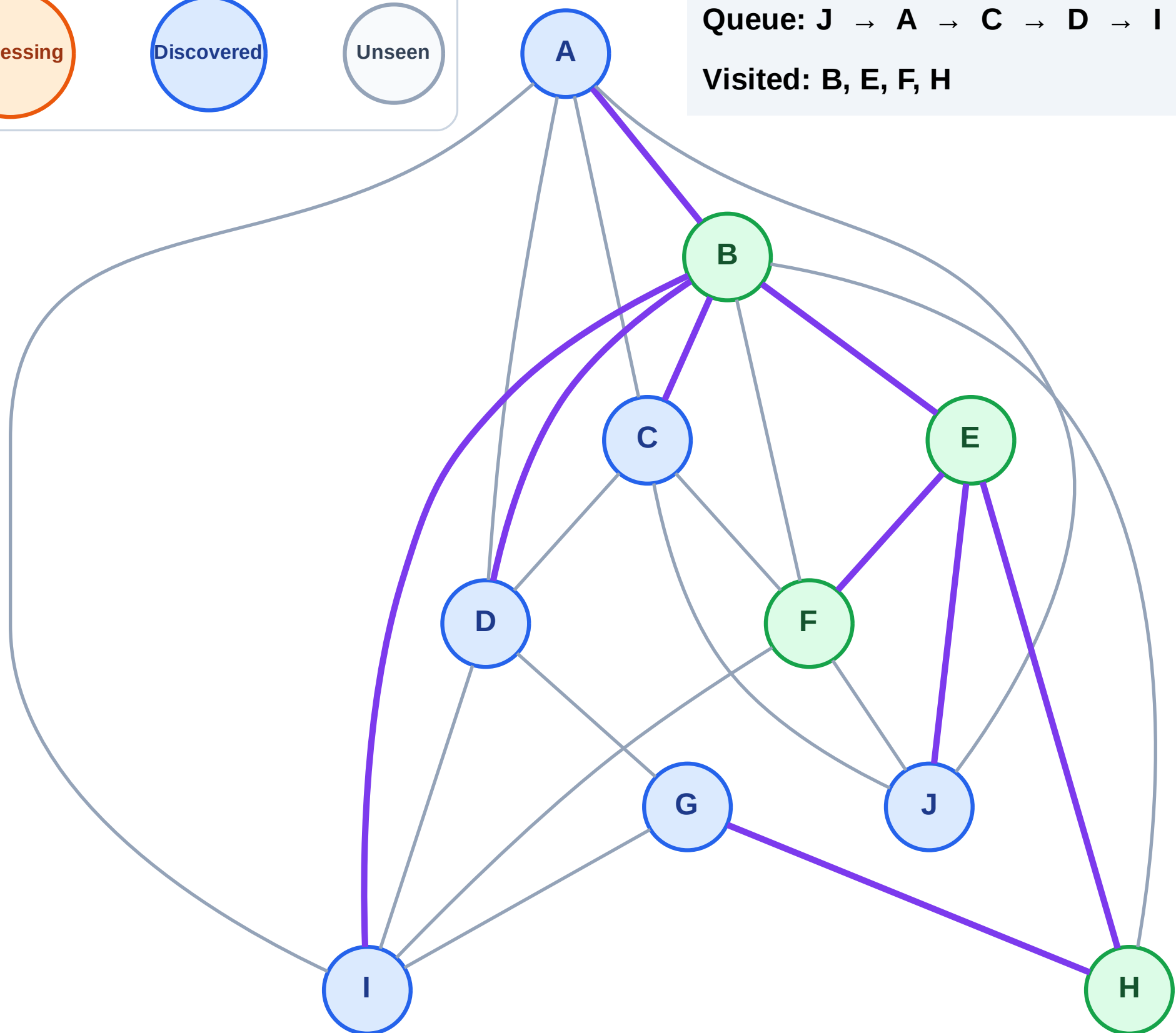
Discovered



Unseen

Queue: J → A → C → D → I → G

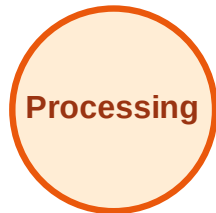
Visited: B, E, F, H



BFS Traversal

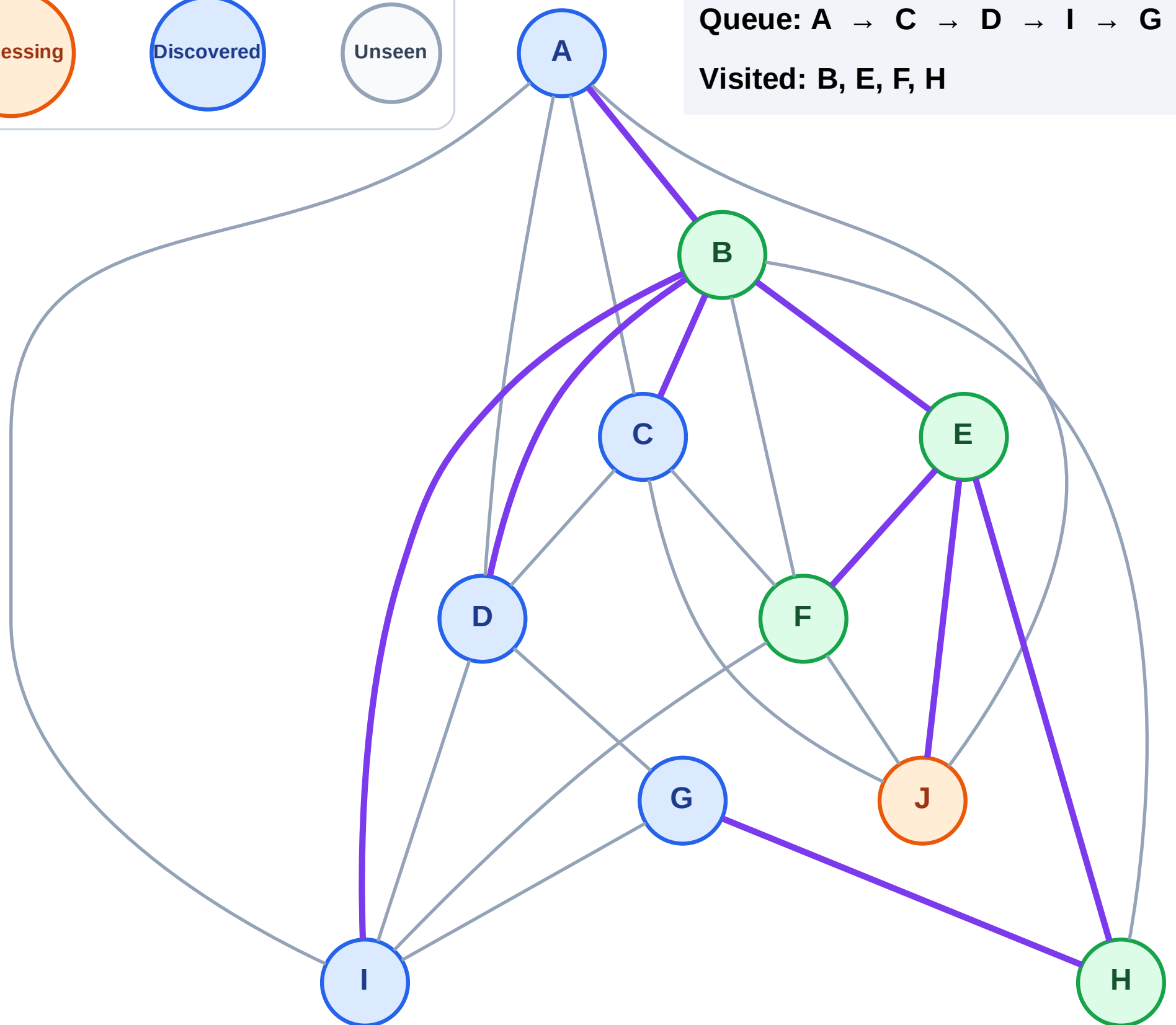
Step 10: Process node J

Legend



Queue: A → C → D → I → G

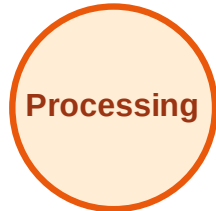
Visited: B, E, F, H



BFS Traversal

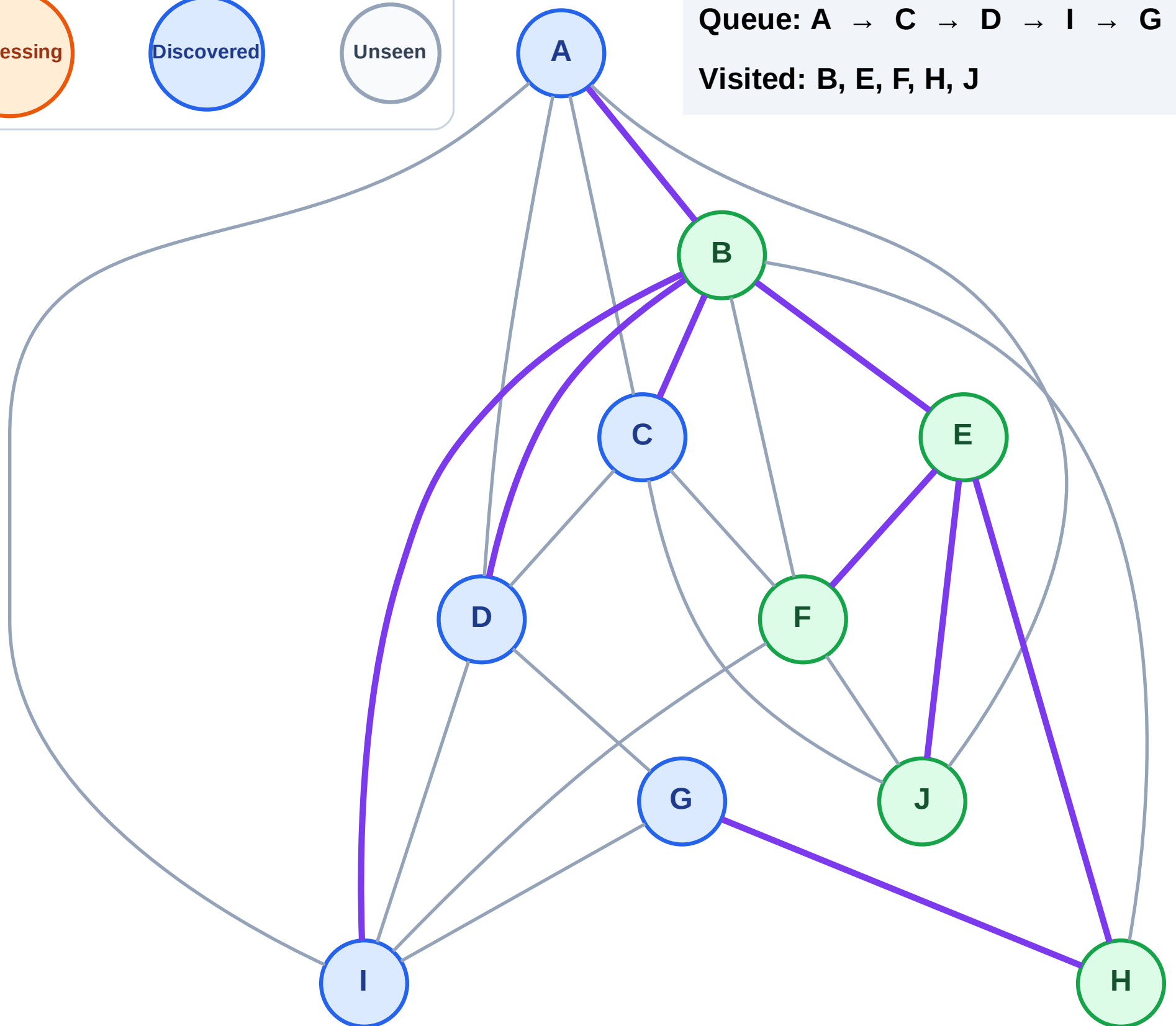
Step 11: No new neighbors from J

Legend



Queue: A → C → D → I → G

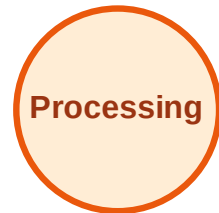
Visited: B, E, F, H, J



BFS Traversal

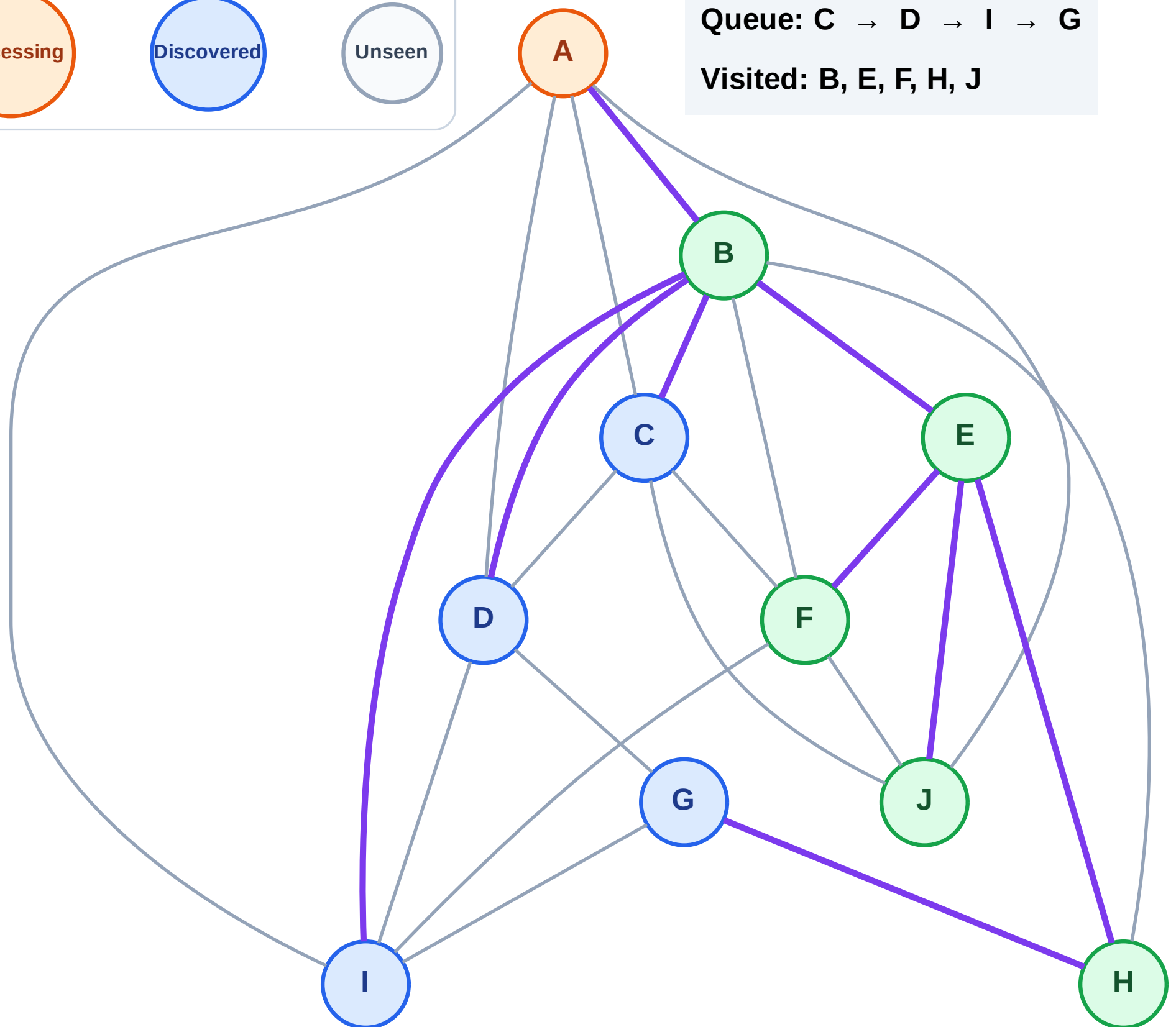
Step 12: Process node A

Legend



Queue: C → D → I → G

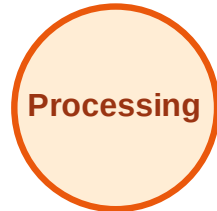
Visited: B, E, F, H, J



BFS Traversal

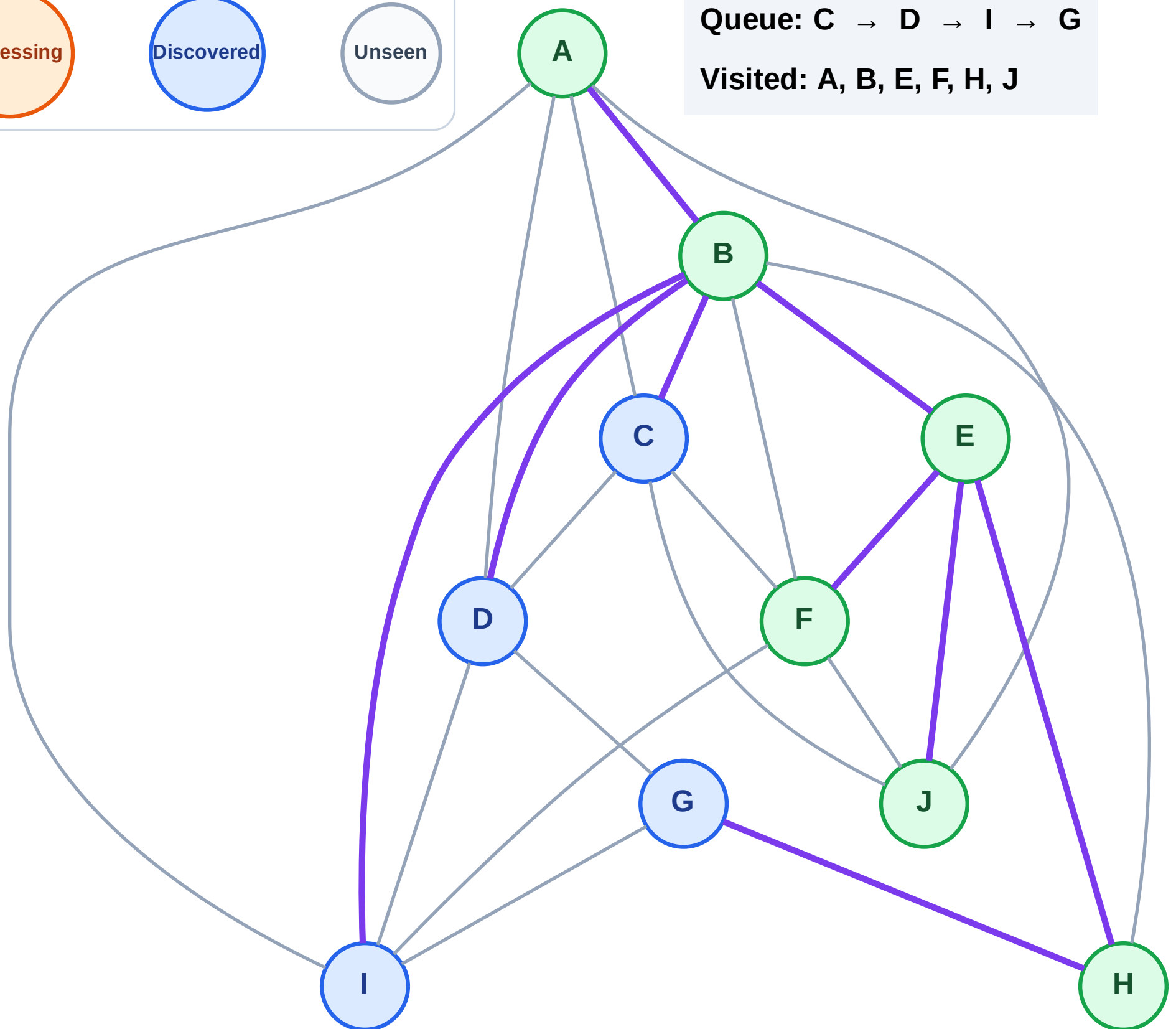
Step 13: No new neighbors from A

Legend



Queue: C → D → I → G

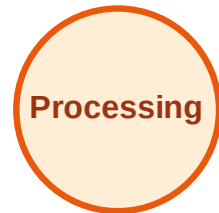
Visited: A, B, E, F, H, J



BFS Traversal

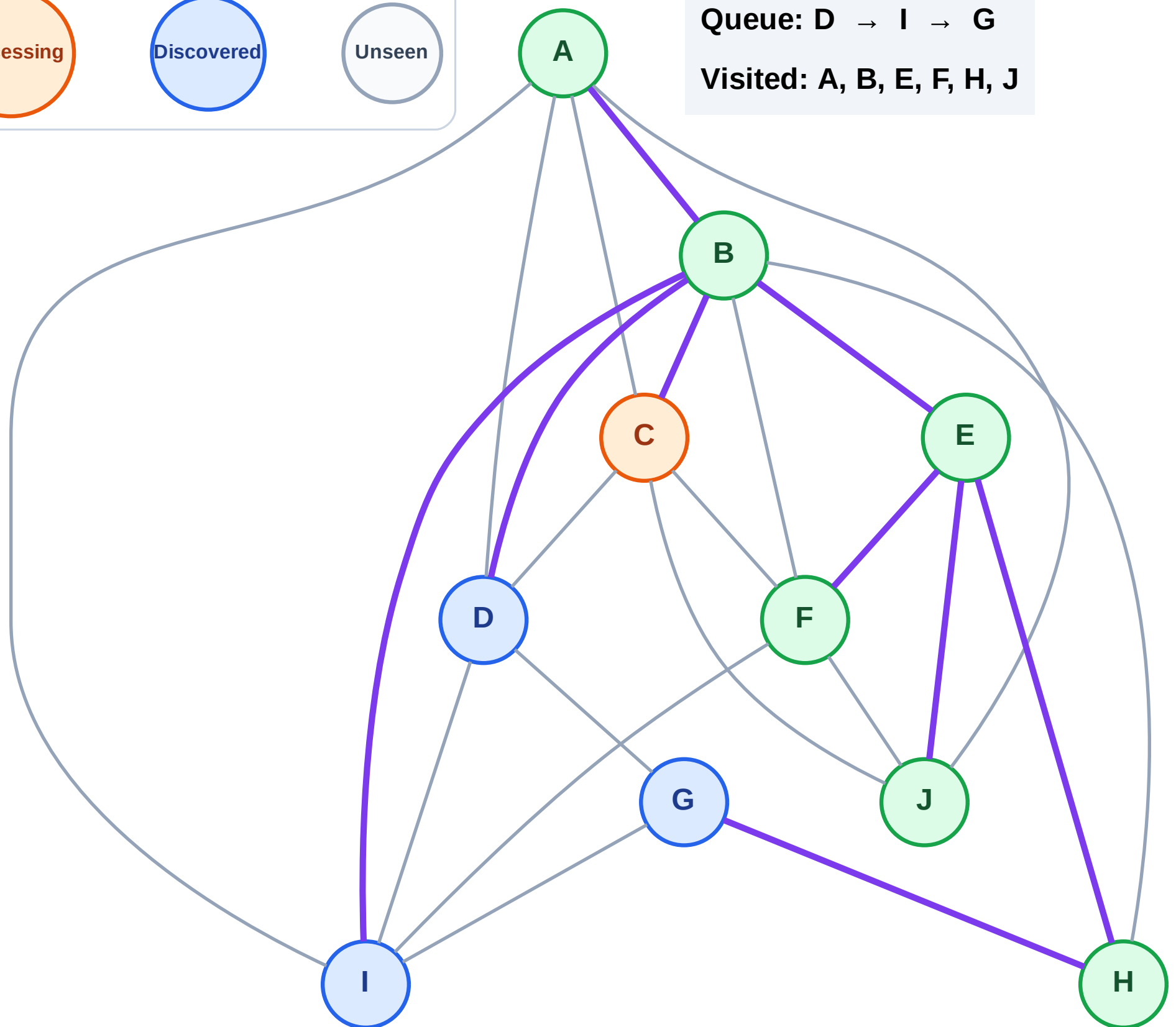
Step 14: Process node C

Legend



Queue: D → I → G

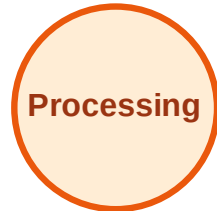
Visited: A, B, E, F, H, J



BFS Traversal

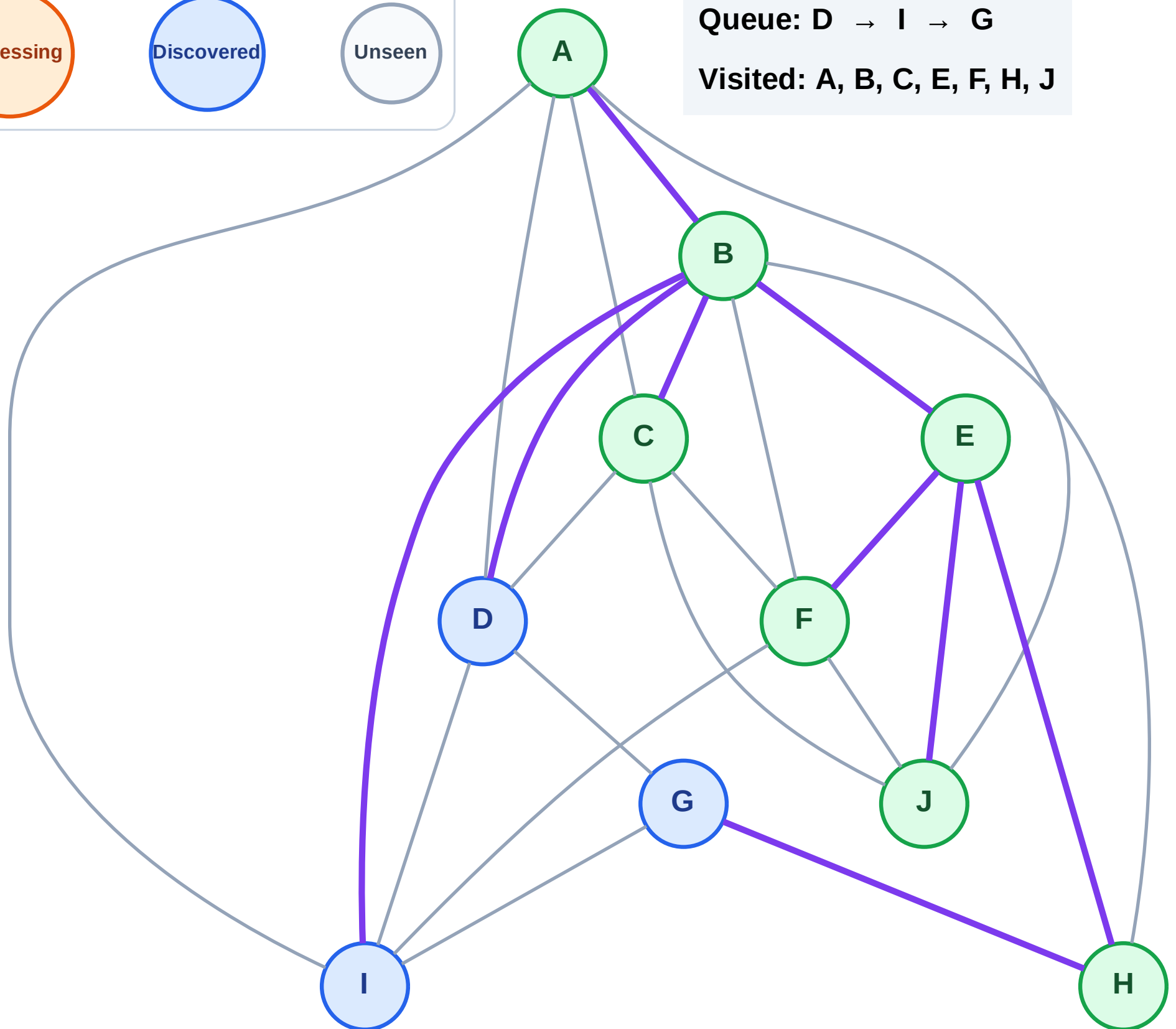
Step 15: No new neighbors from C

Legend



Queue: D → I → G

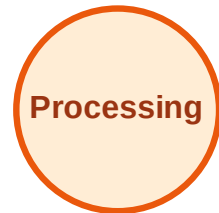
Visited: A, B, C, E, F, H, J



BFS Traversal

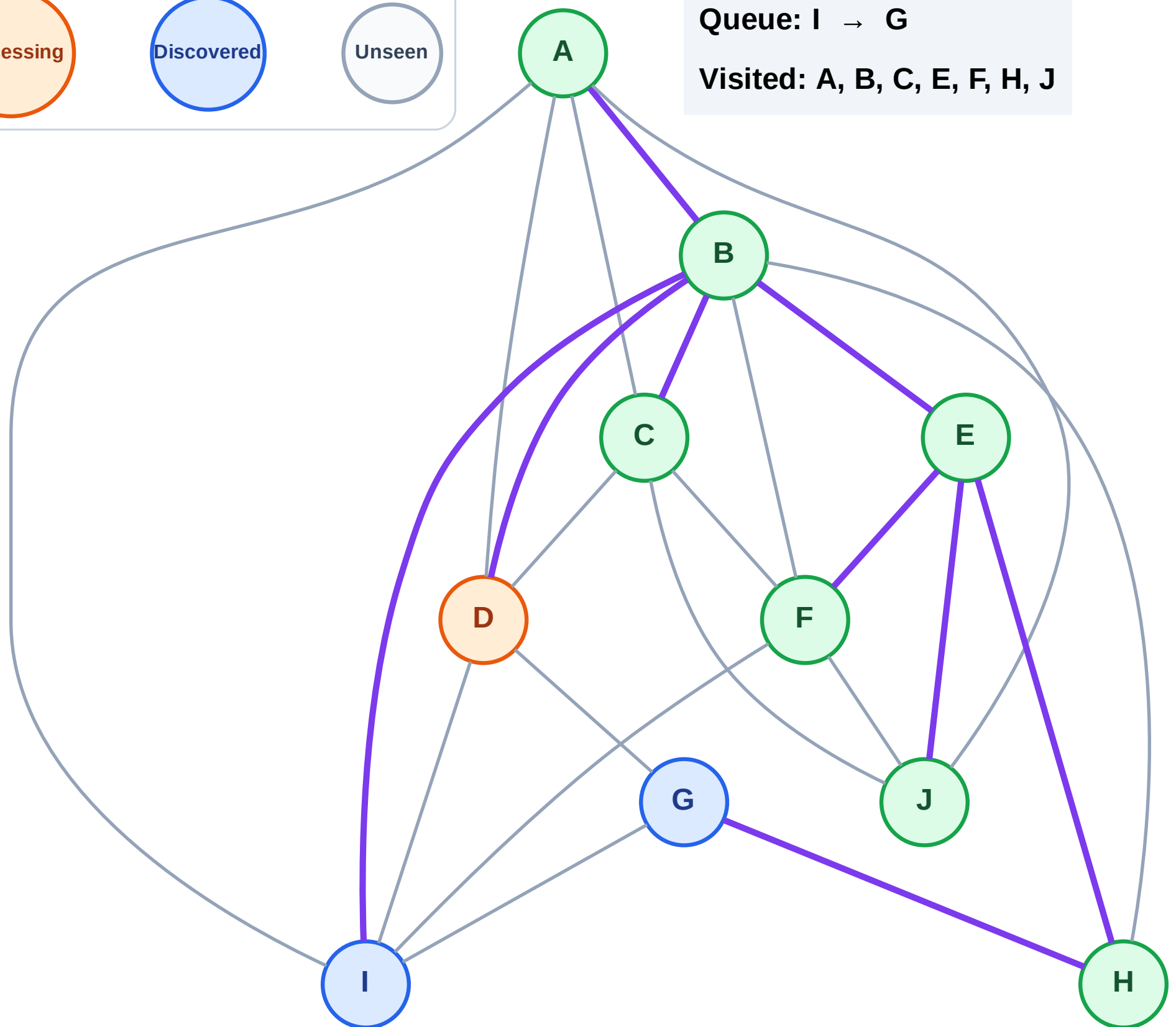
Step 16: Process node D

Legend



Queue: I → G

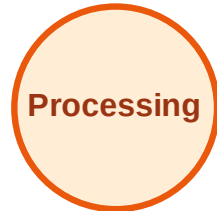
Visited: A, B, C, E, F, H, J



BFS Traversal

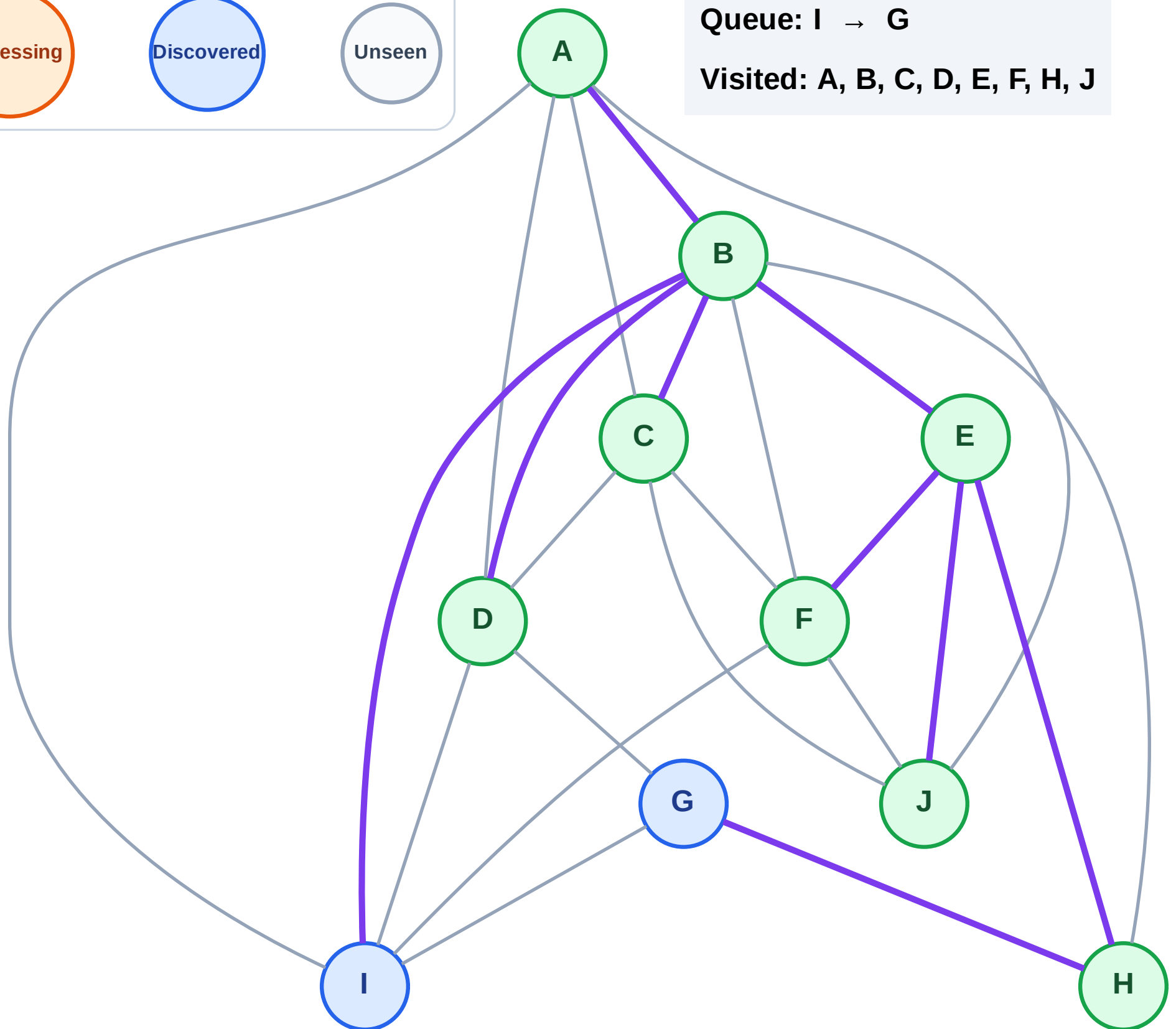
Step 17: No new neighbors from D

Legend



Queue: I → G

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

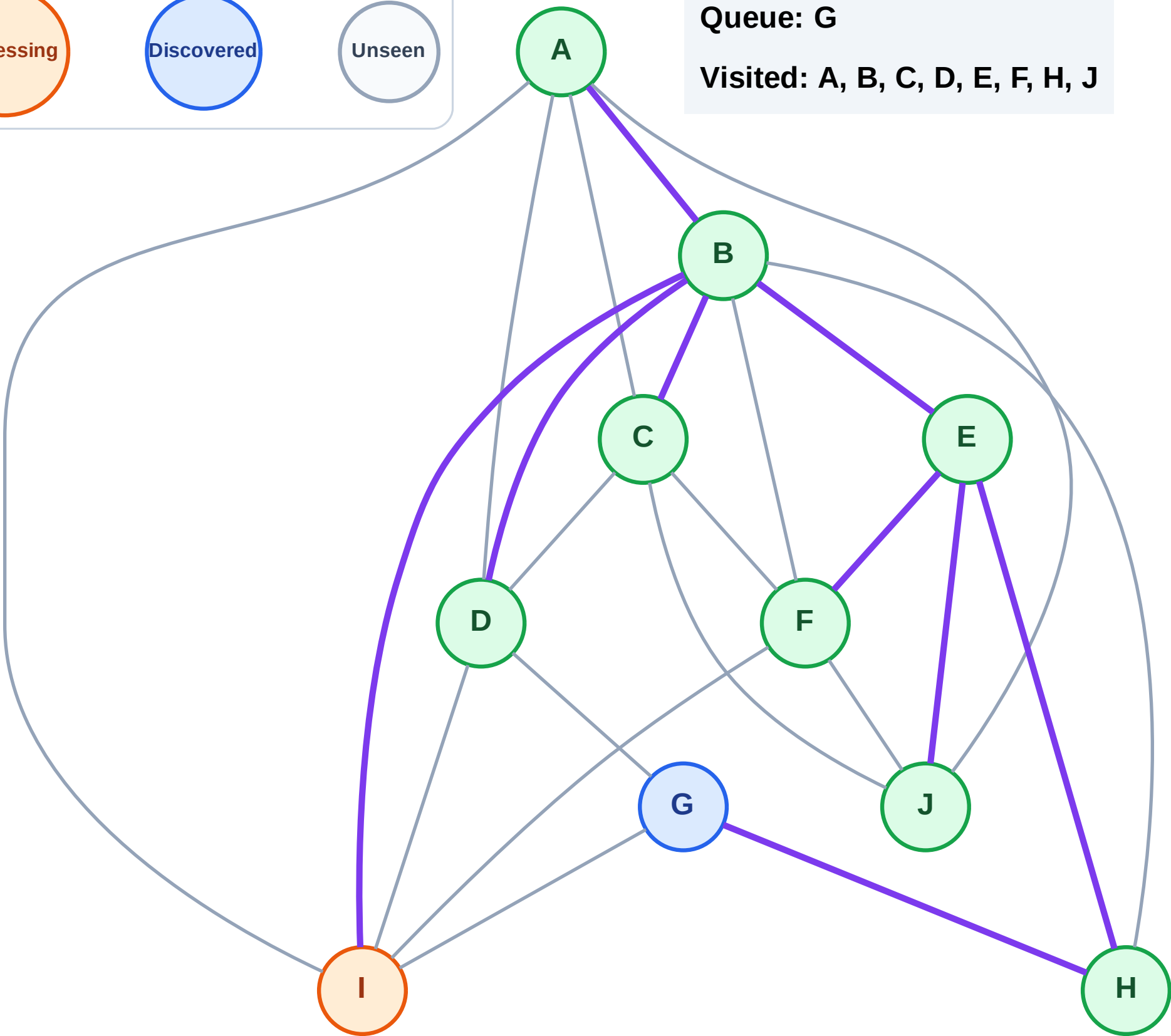
Processing

Discovered

Unseen

Queue: G

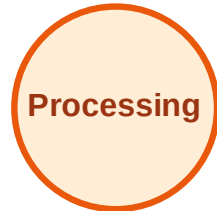
Visited: A, B, C, D, E, F, H, J



BFS Traversal

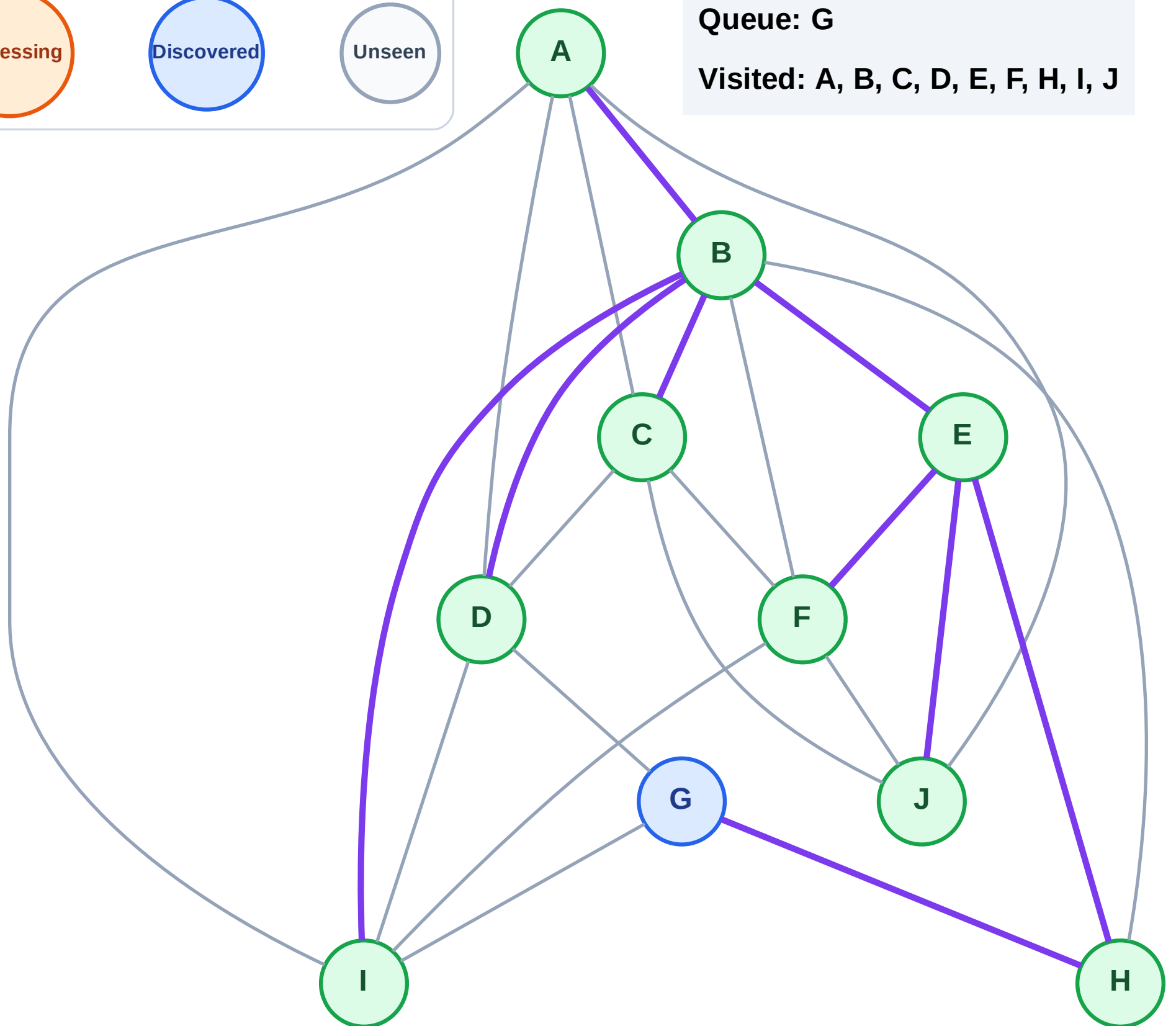
Step 19: No new neighbors from I

Legend



Queue: G

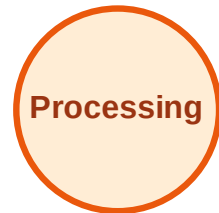
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

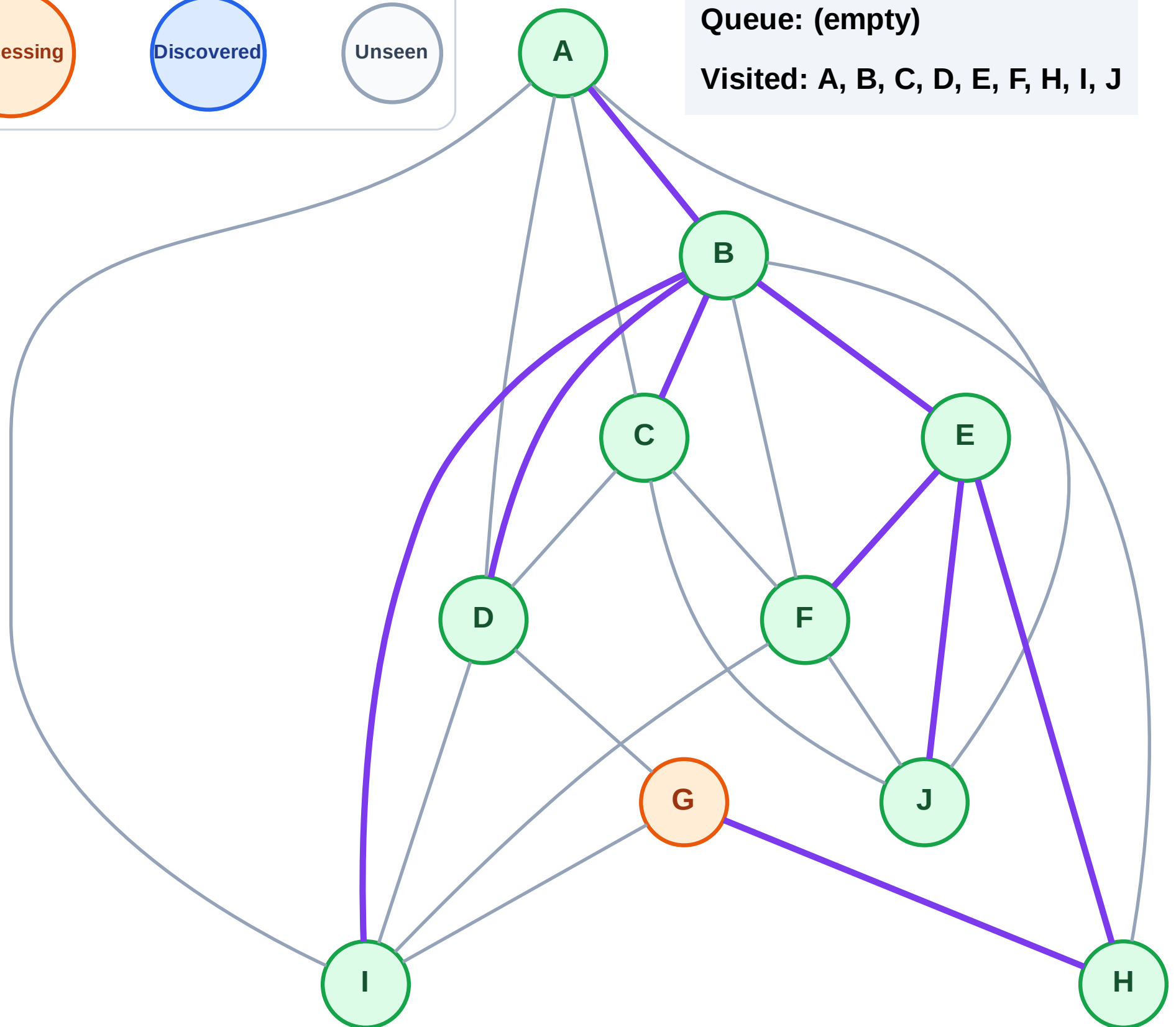
Step 20: Process node G

Legend



Queue: (empty)

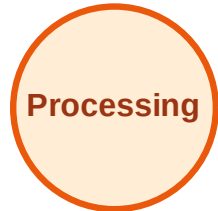
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

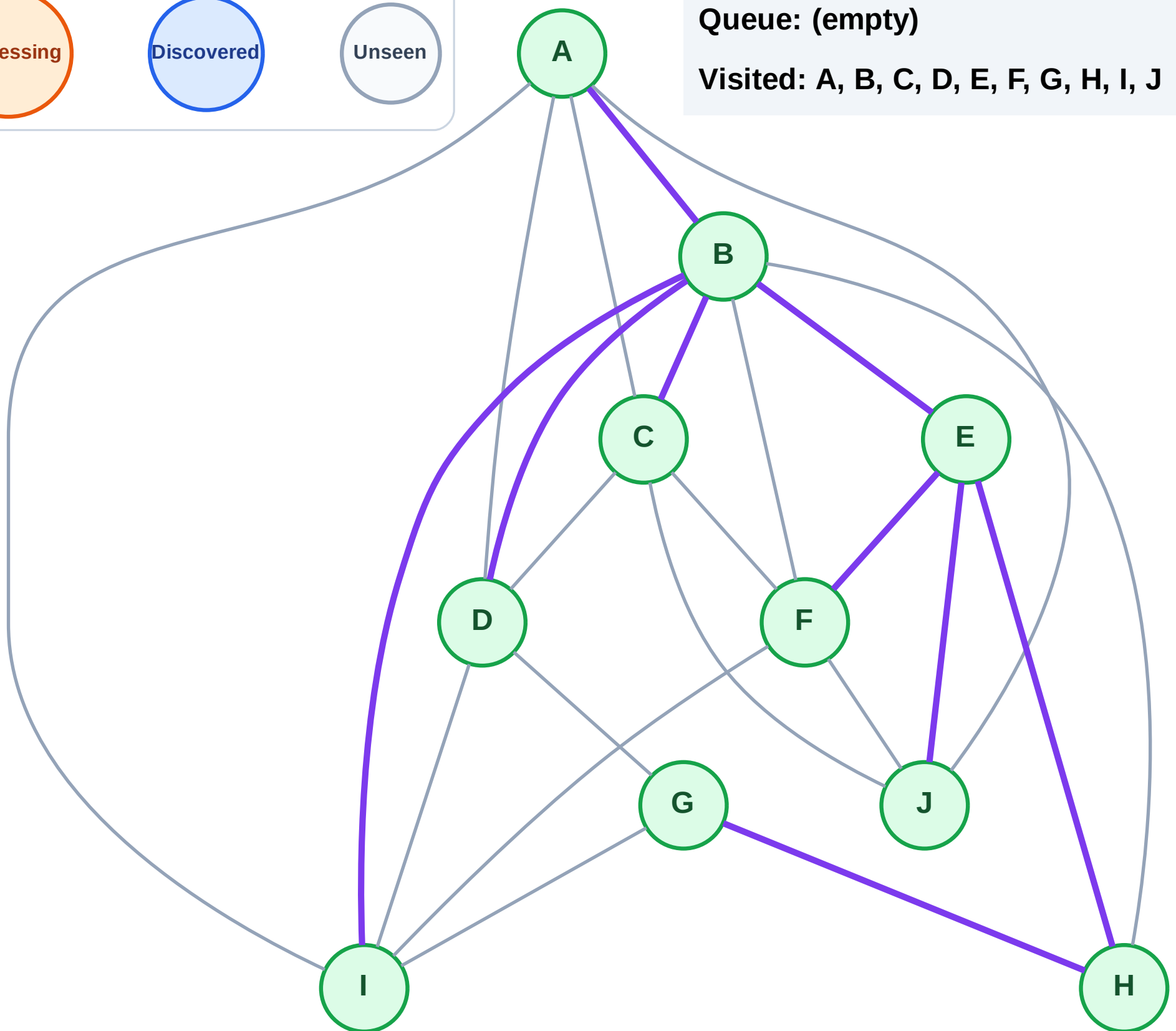
Step 21: No new neighbors from G

Legend



Queue: (empty)

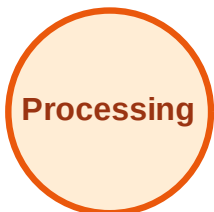
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

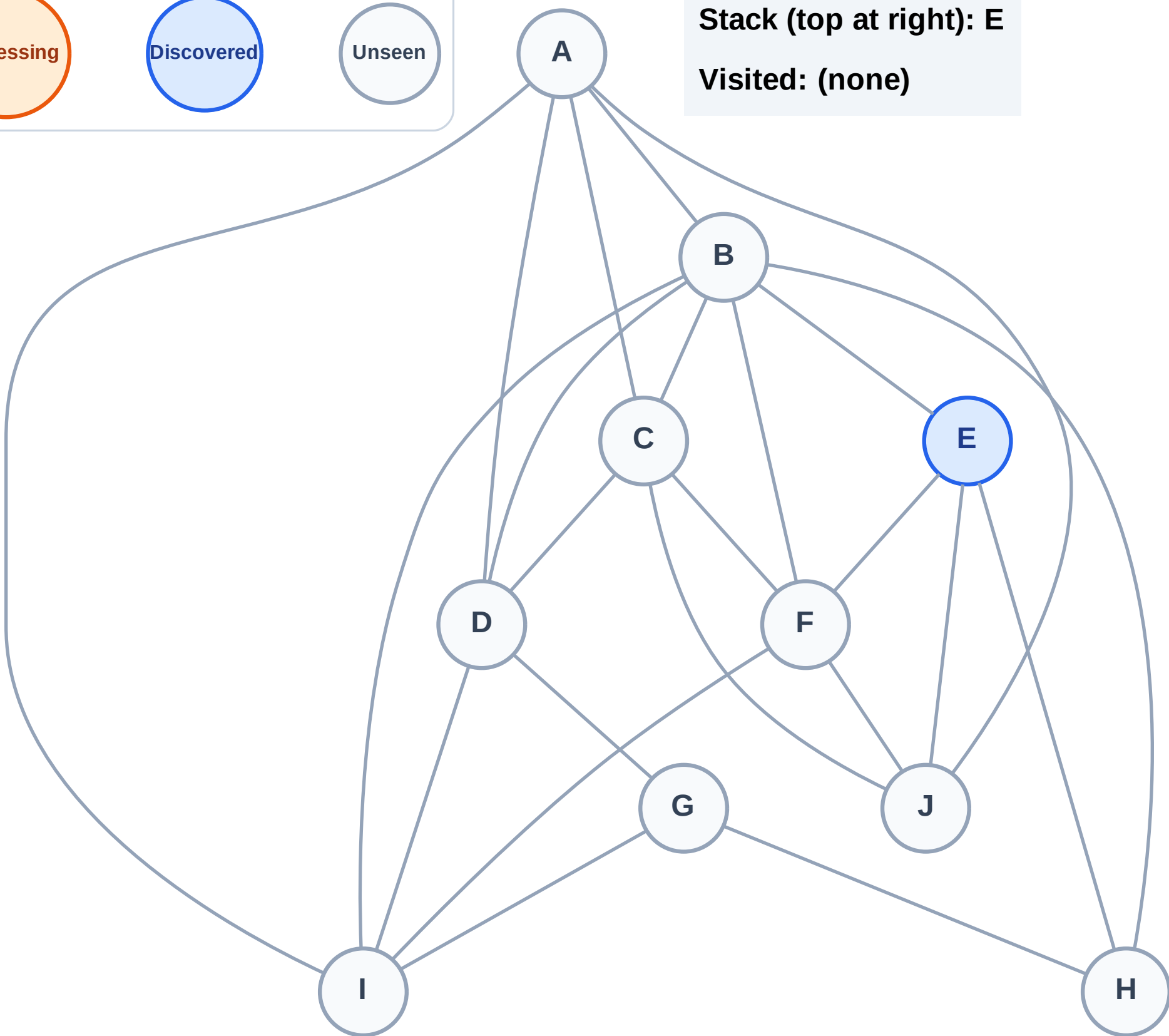
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

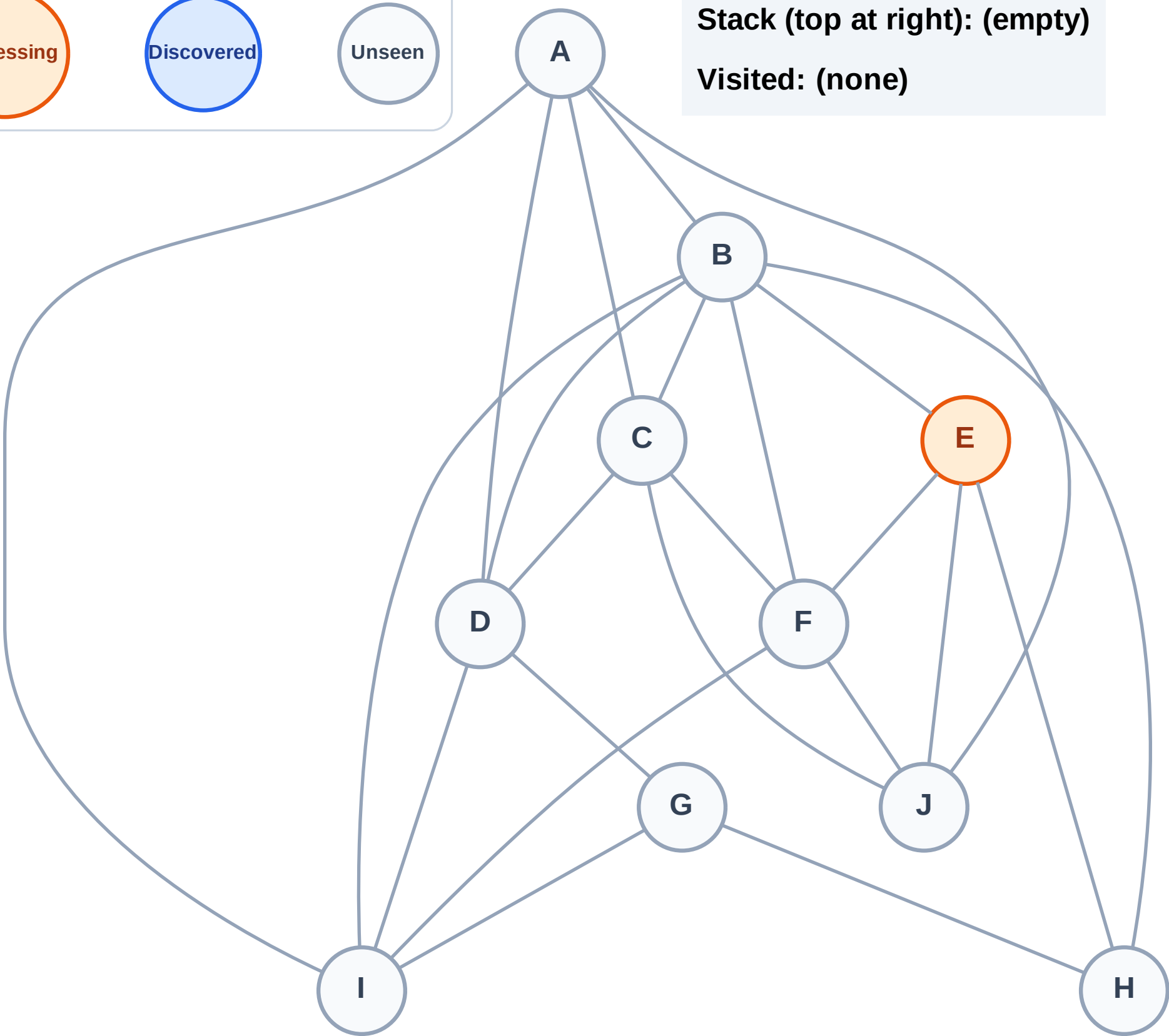
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, H, F, B from E

Legend

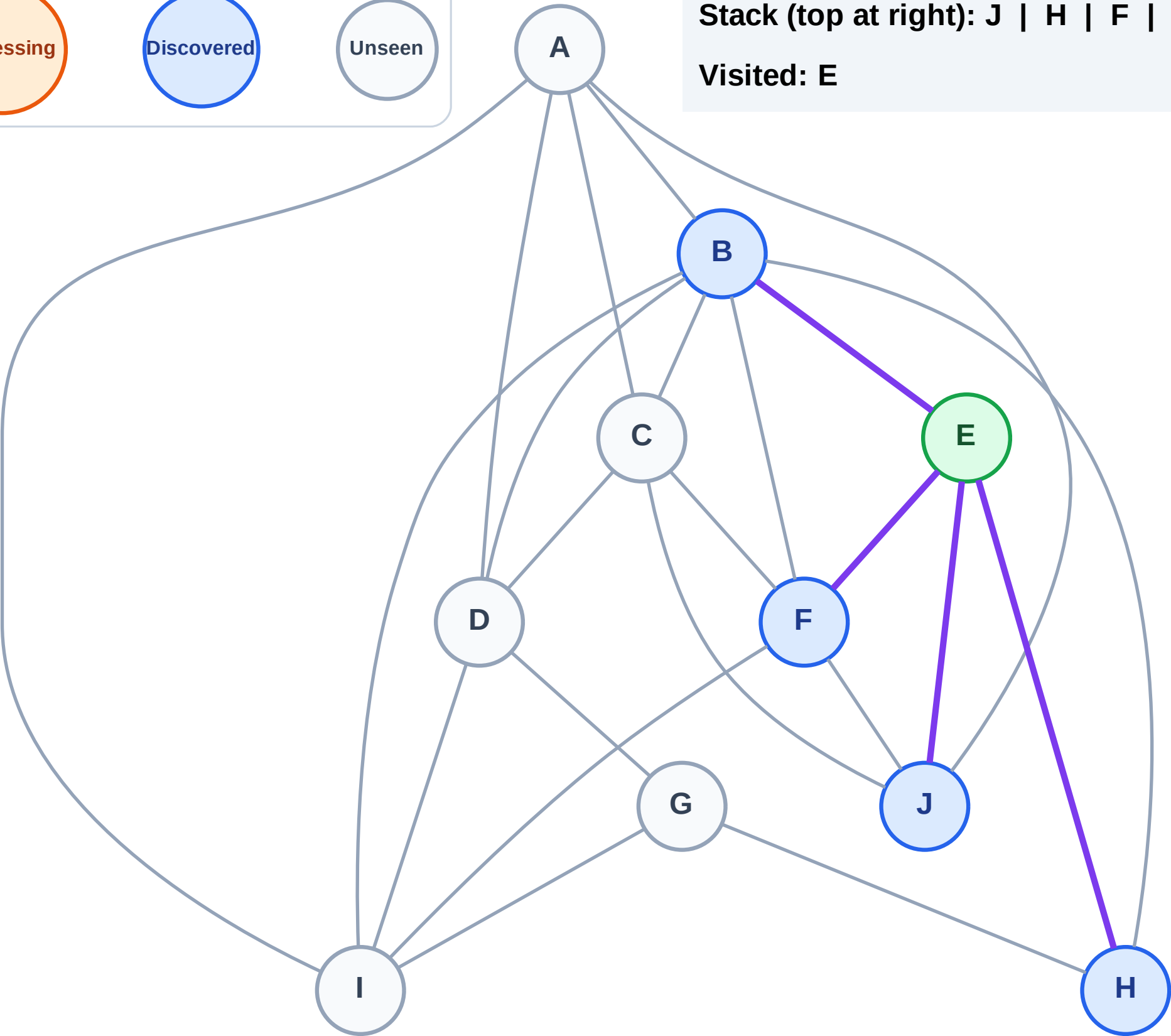
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | F | B
Visited: E



DFS Traversal

Step 4: Process node B

Legend

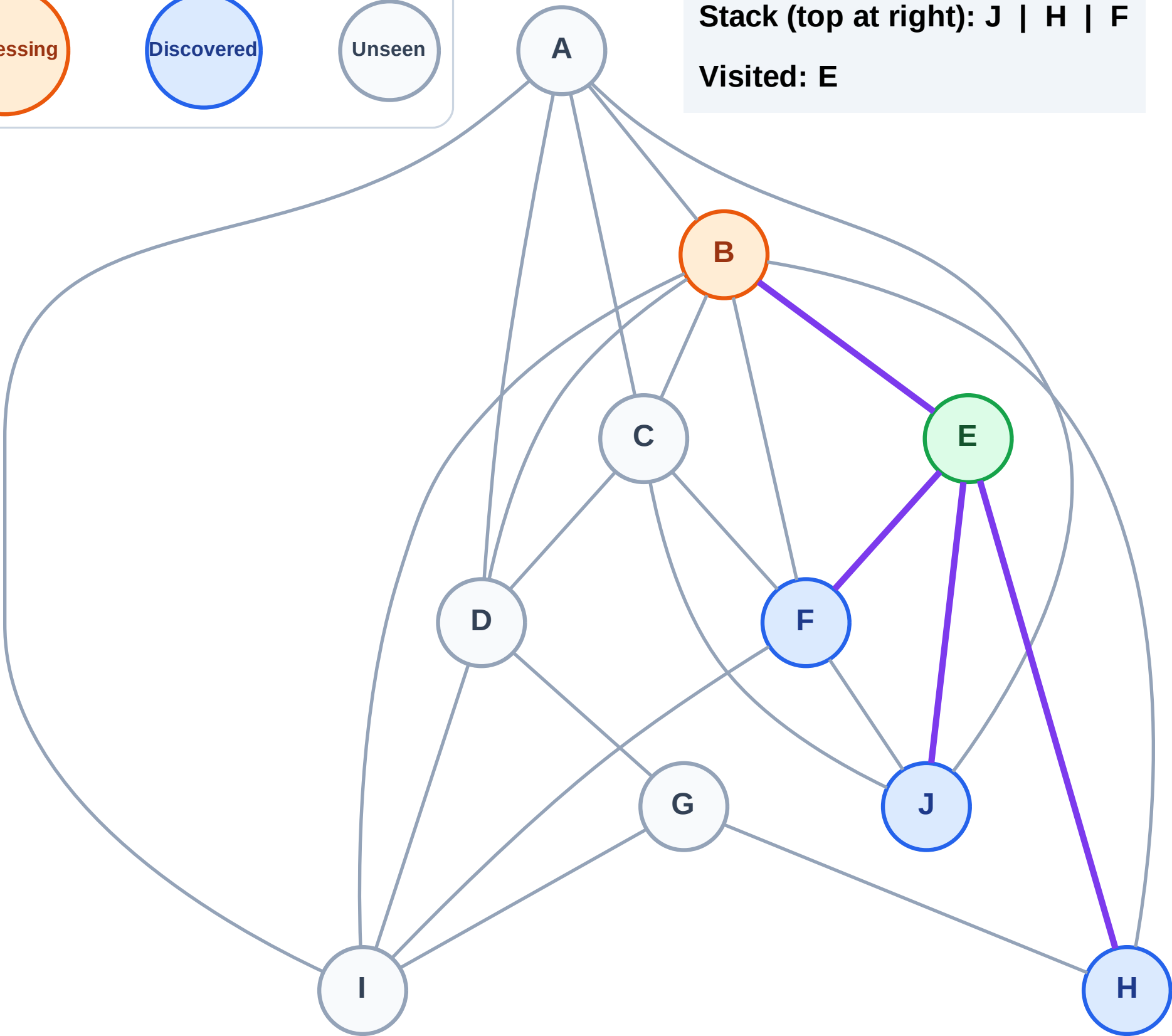
Complete

Processing

Discovered

Unseen

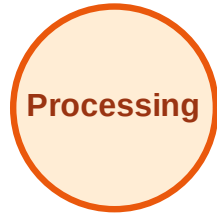
Stack (top at right): J | H | F
Visited: E



DFS Traversal

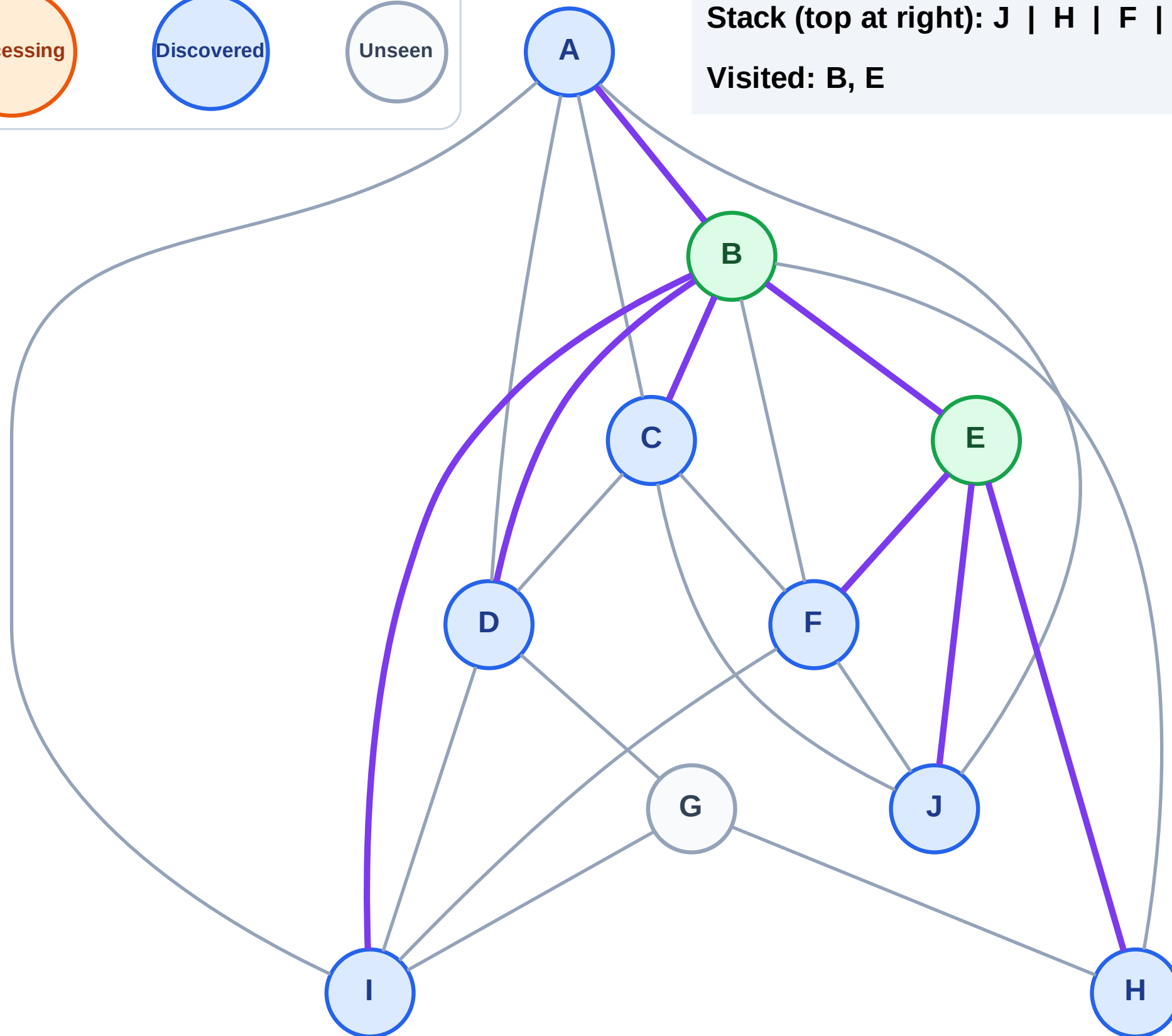
Step 5: Discover I, D, C, A from B

Legend



Stack (top at right): J | H | F | I | D | C | A

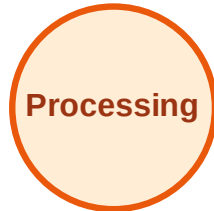
Visited: B, E



DFS Traversal

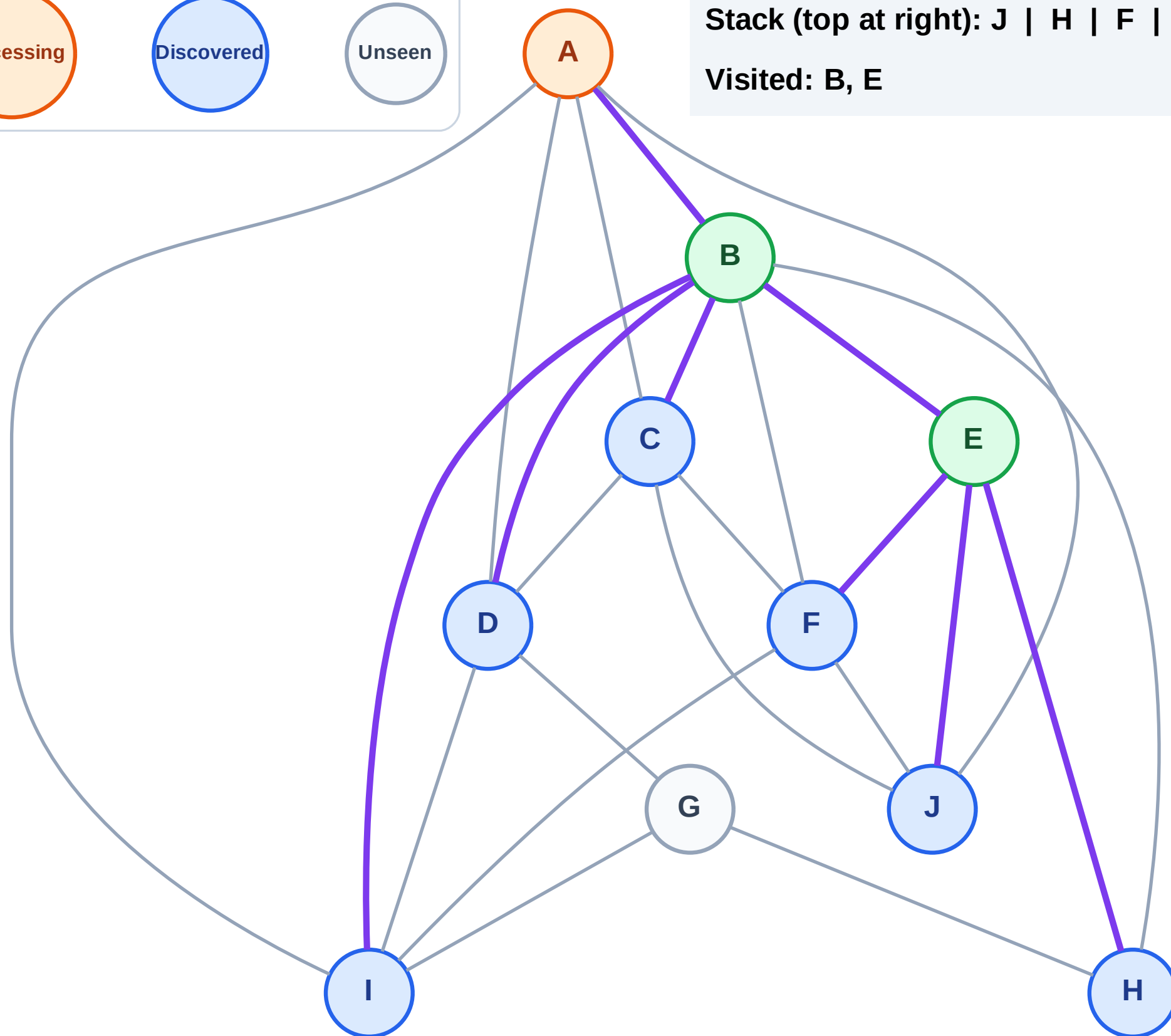
Step 6: Process node A

Legend



Stack (top at right): J | H | F | I | D | C

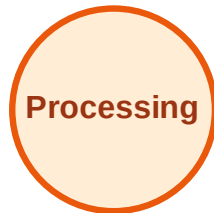
Visited: B, E



DFS Traversal

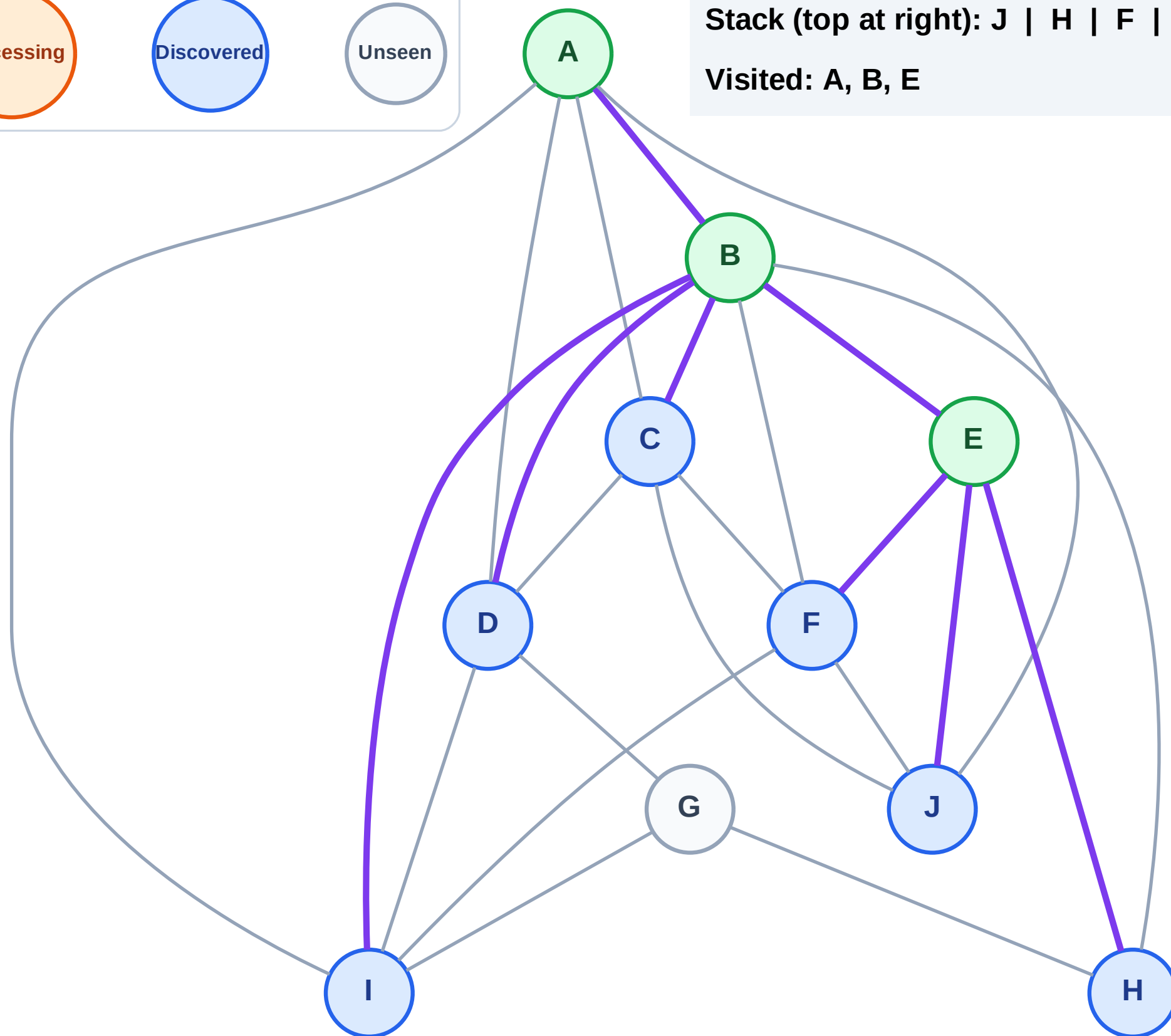
Step 7: No new neighbors from A

Legend



Stack (top at right): J | H | F | I | D | C

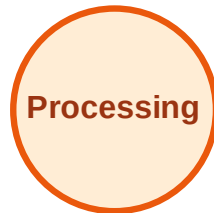
Visited: A, B, E



DFS Traversal

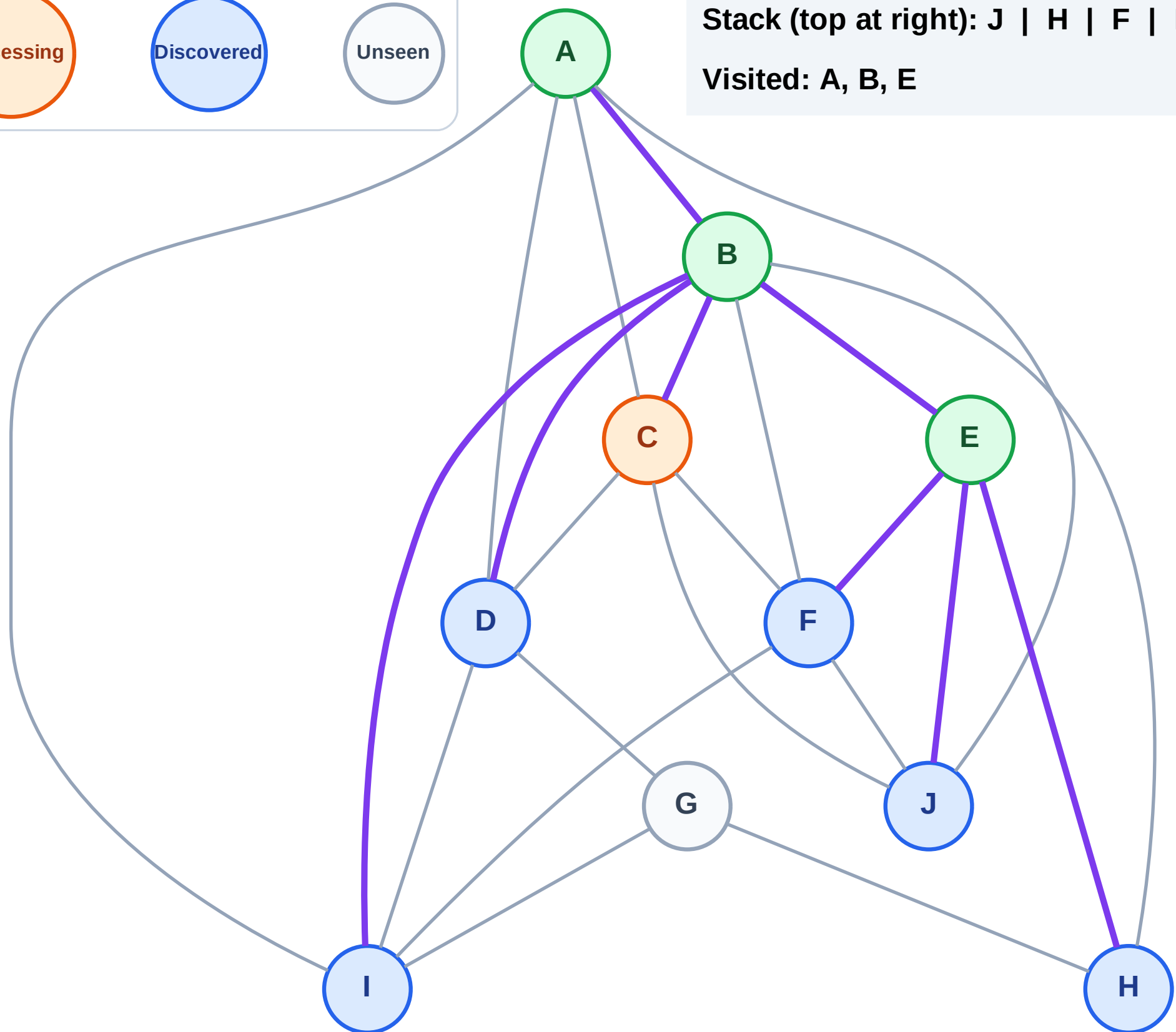
Step 8: Process node C

Legend



Stack (top at right): J | H | F | I | D

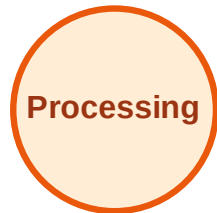
Visited: A, B, E



DFS Traversal

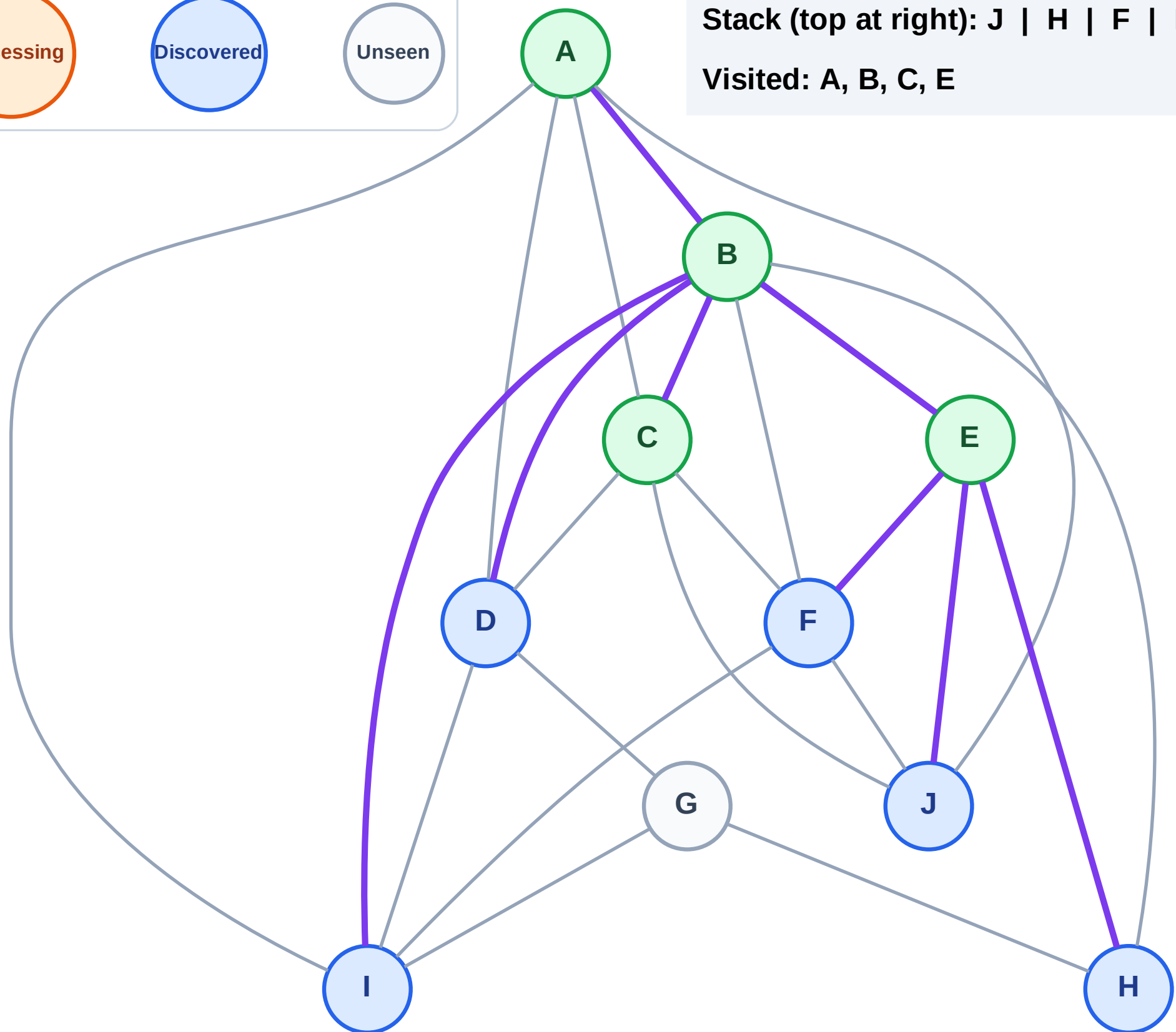
Step 9: No new neighbors from C

Legend



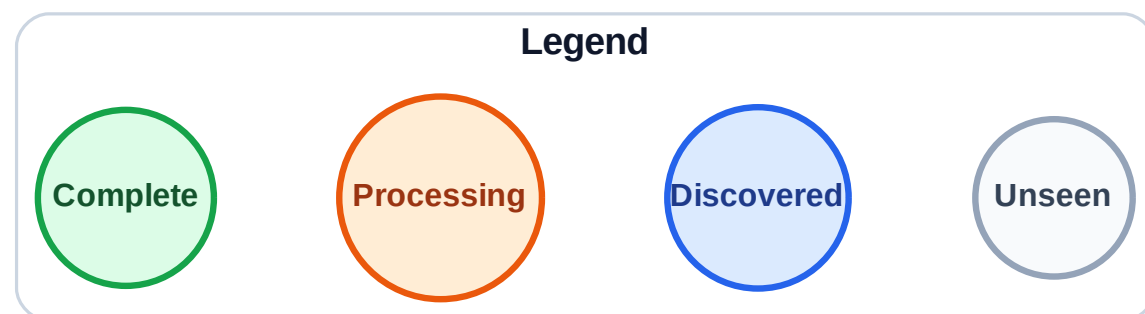
Stack (top at right): J | H | F | I | D

Visited: A, B, C, E

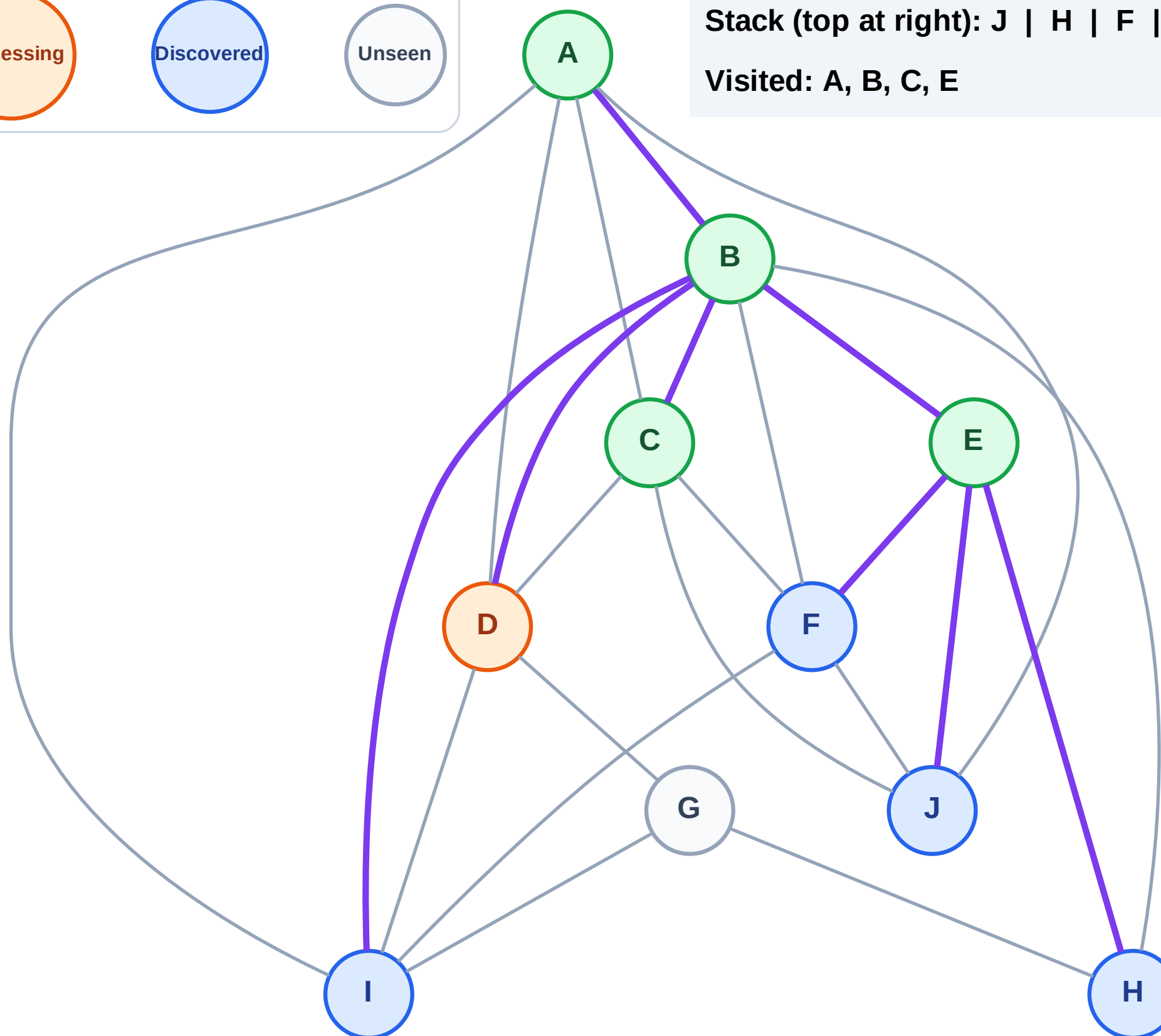


DFS Traversal

Step 10: Process node D



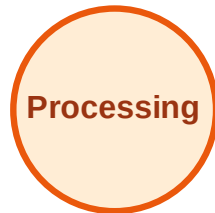
Stack (top at right): J | H | F | I
Visited: A, B, C, E



DFS Traversal

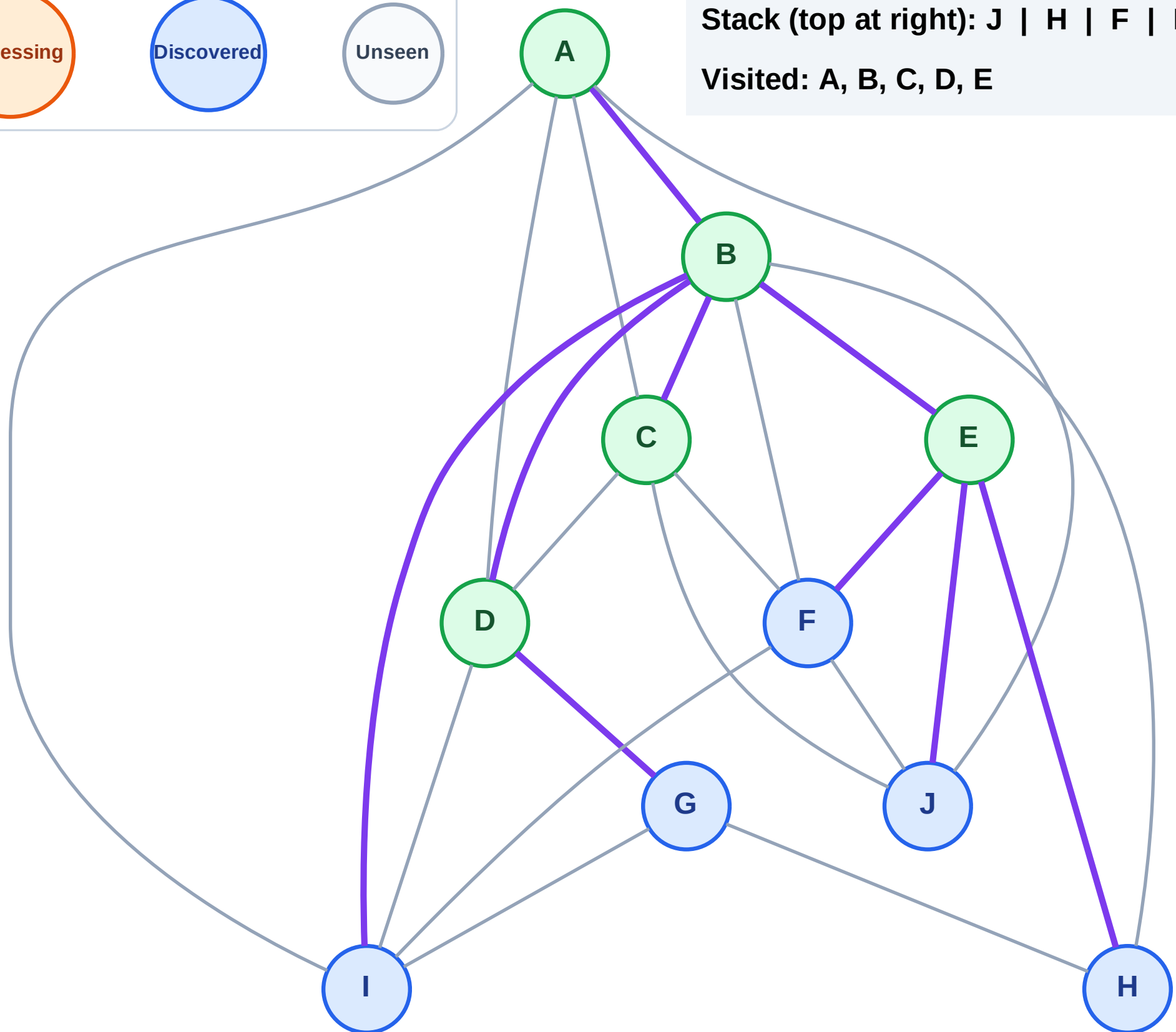
Step 11: Discover G from D

Legend



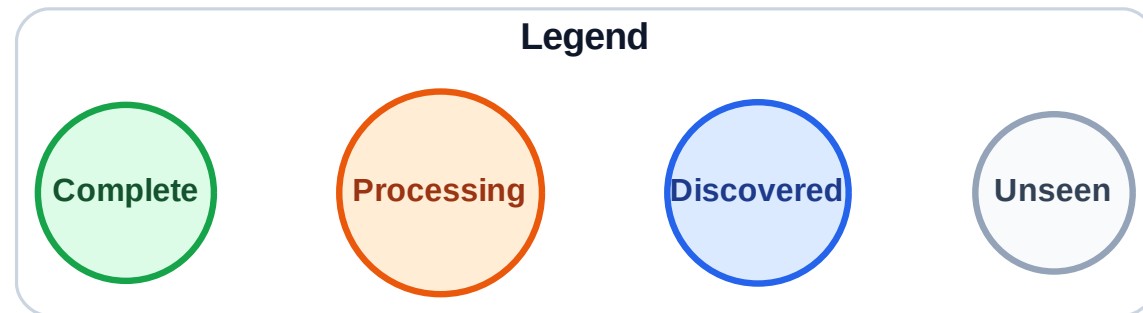
Stack (top at right): J | H | F | I | G

Visited: A, B, C, D, E

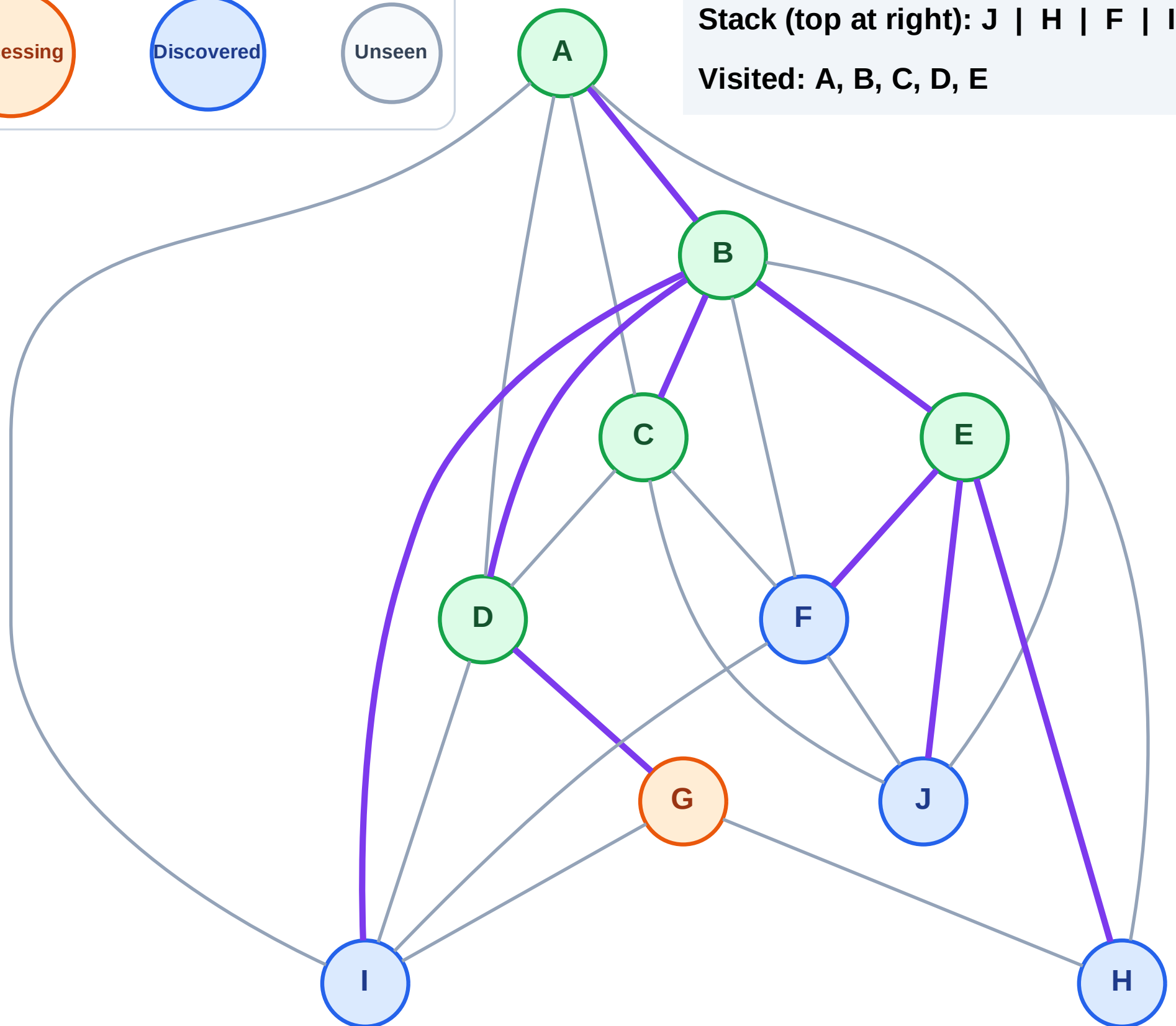


DFS Traversal

Step 12: Process node G



Stack (top at right): J | H | F | I
Visited: A, B, C, D, E



DFS Traversal

Step 13: No new neighbors from G

Legend

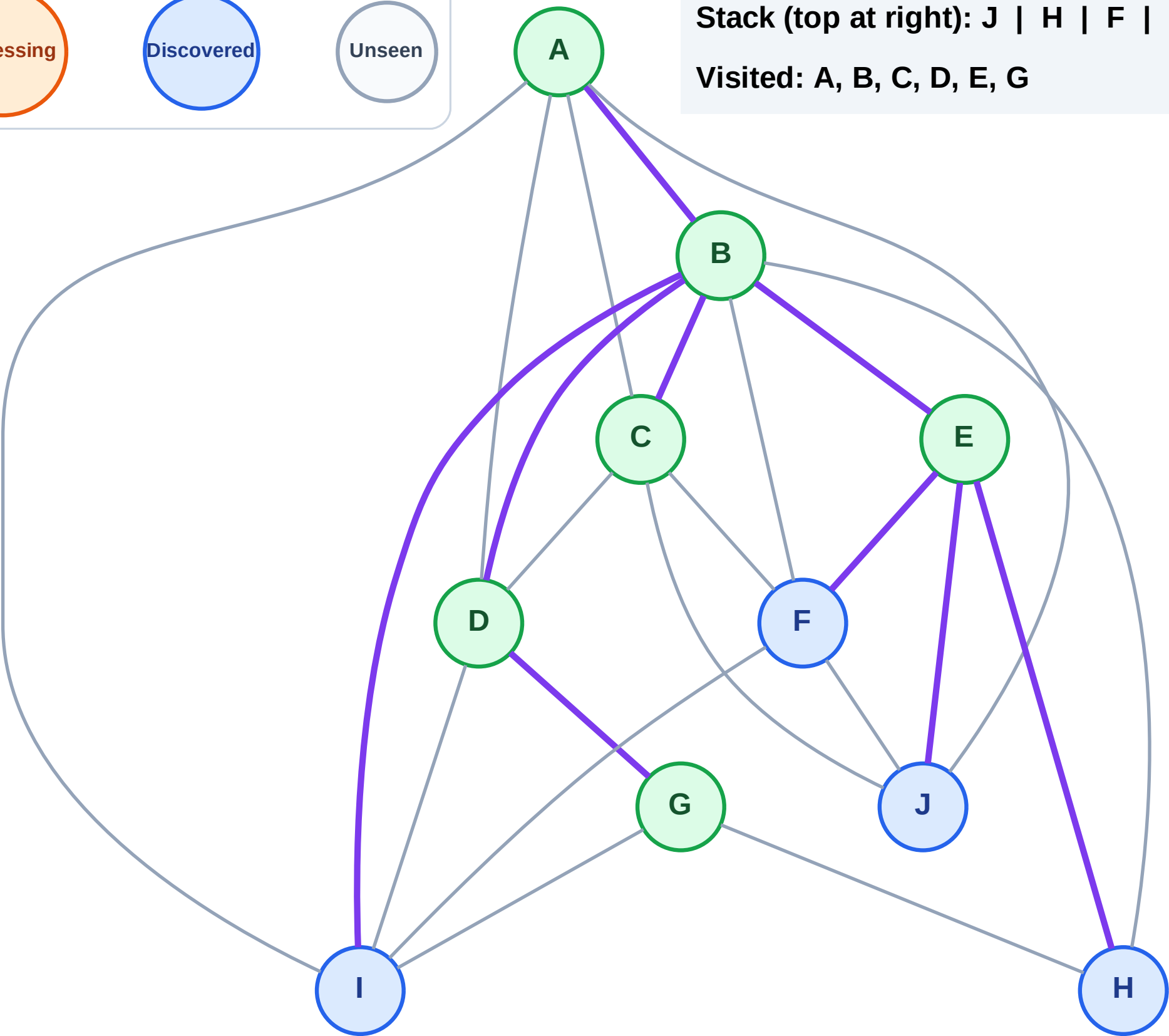
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | F | I
Visited: A, B, C, D, E, G



DFS Traversal

Step 14: Process node I

Legend

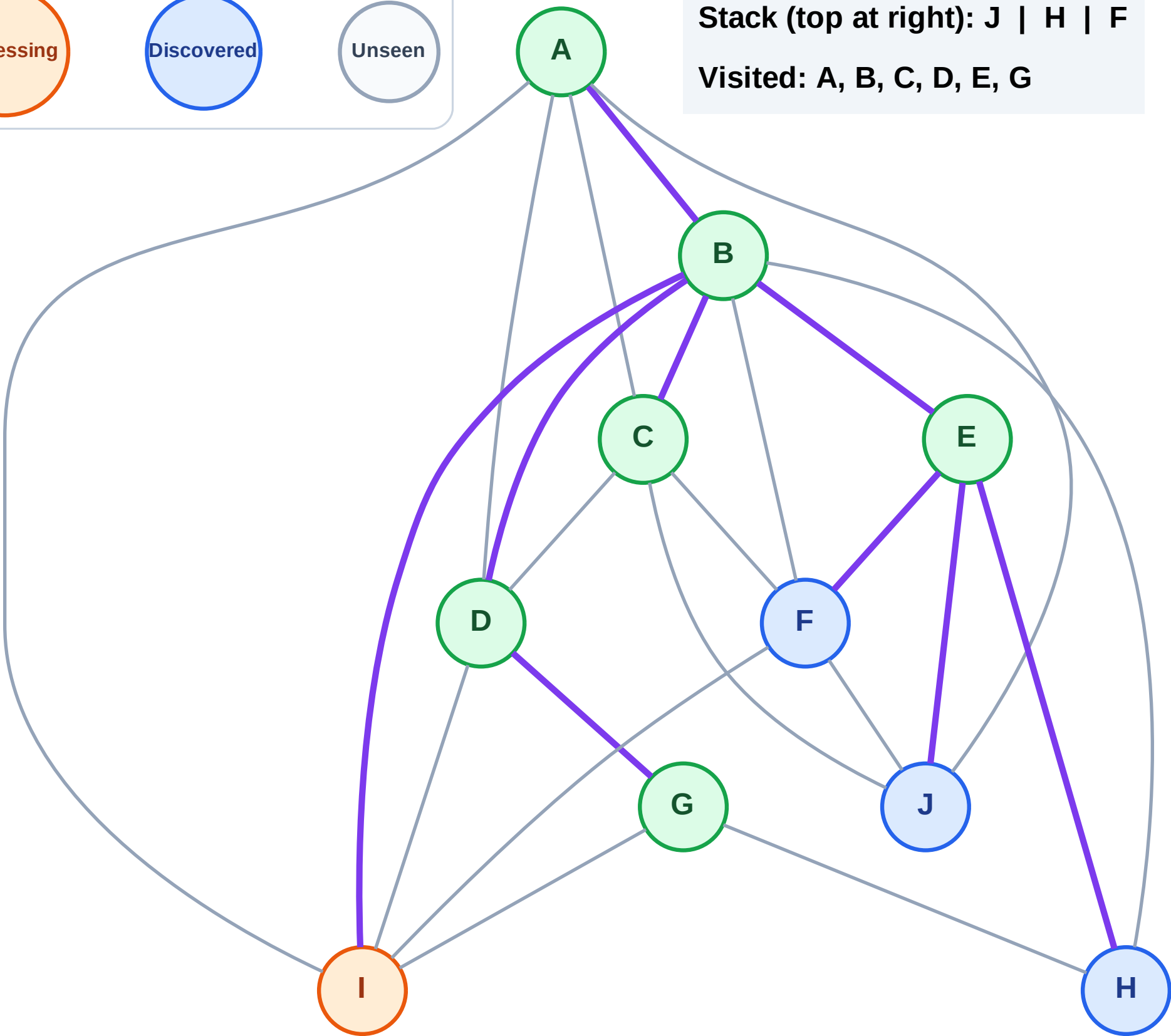
Complete

Processing

Discovered

Unseen

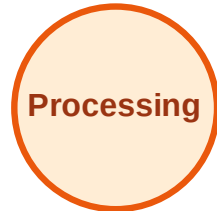
Stack (top at right): J | H | F
Visited: A, B, C, D, E, G



DFS Traversal

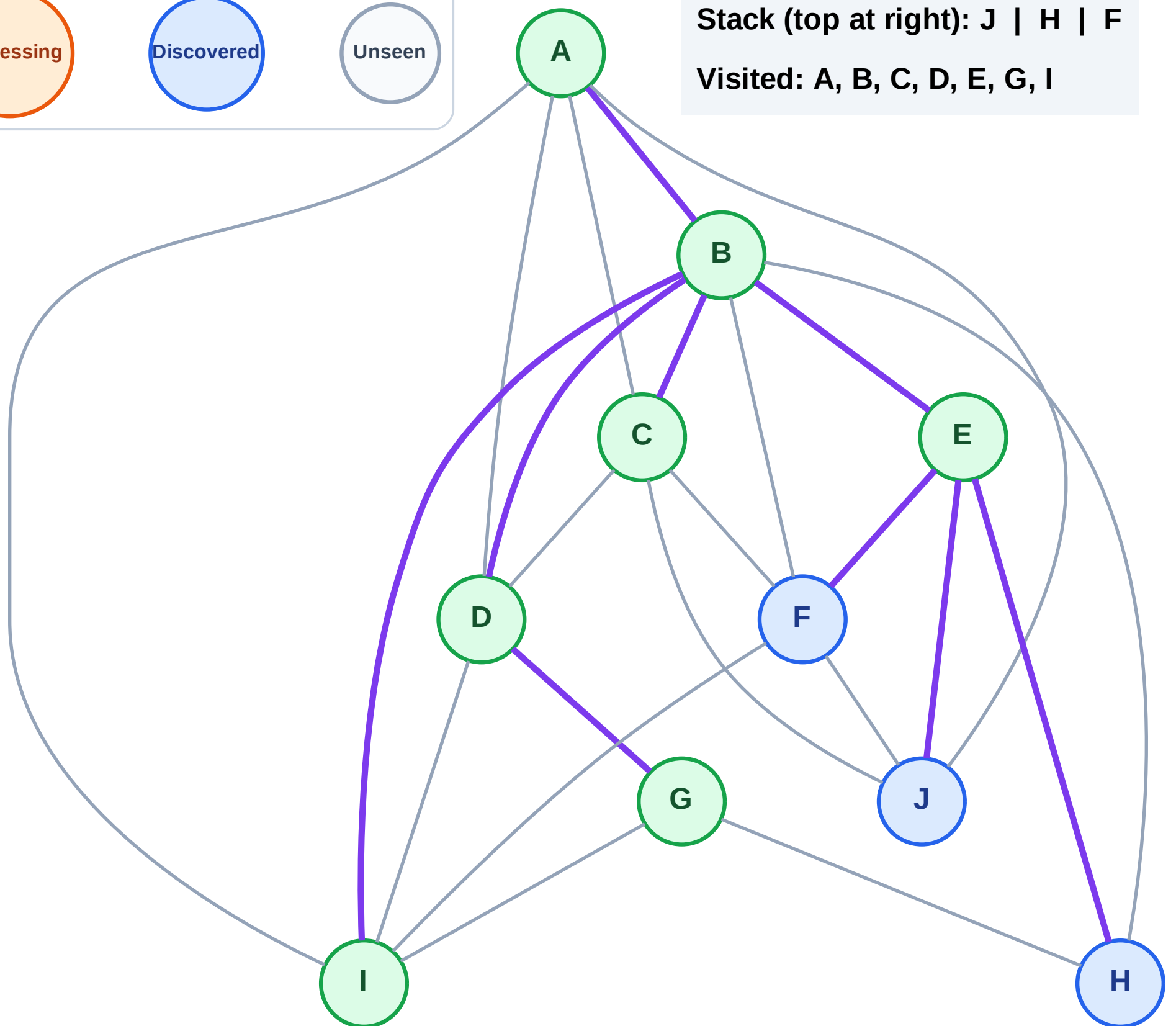
Step 15: No new neighbors from I

Legend



Stack (top at right): J | H | F

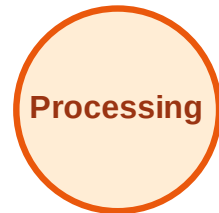
Visited: A, B, C, D, E, G, I



DFS Traversal

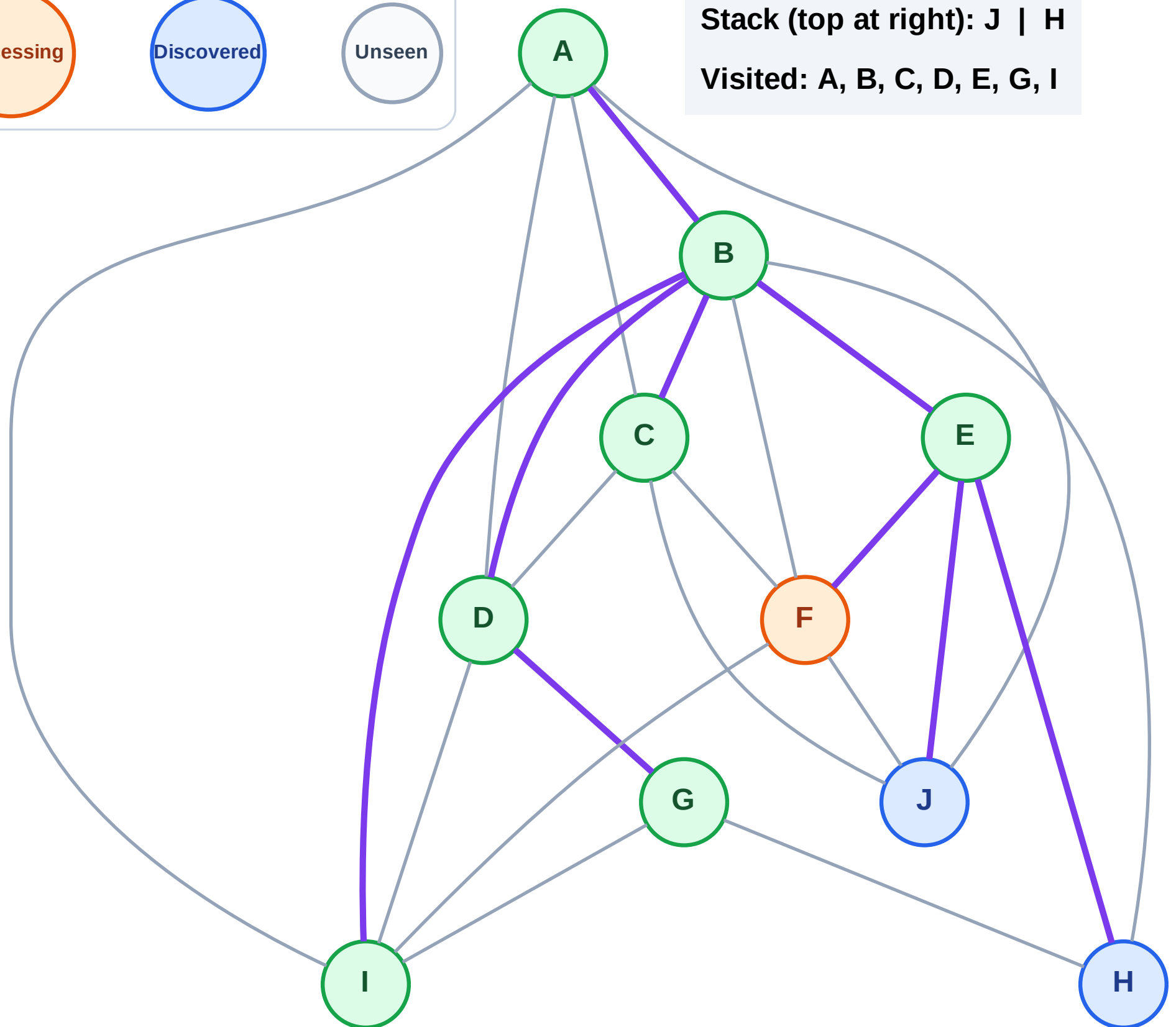
Step 16: Process node F

Legend



Stack (top at right): J | H

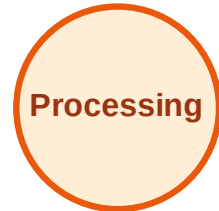
Visited: A, B, C, D, E, G, I



DFS Traversal

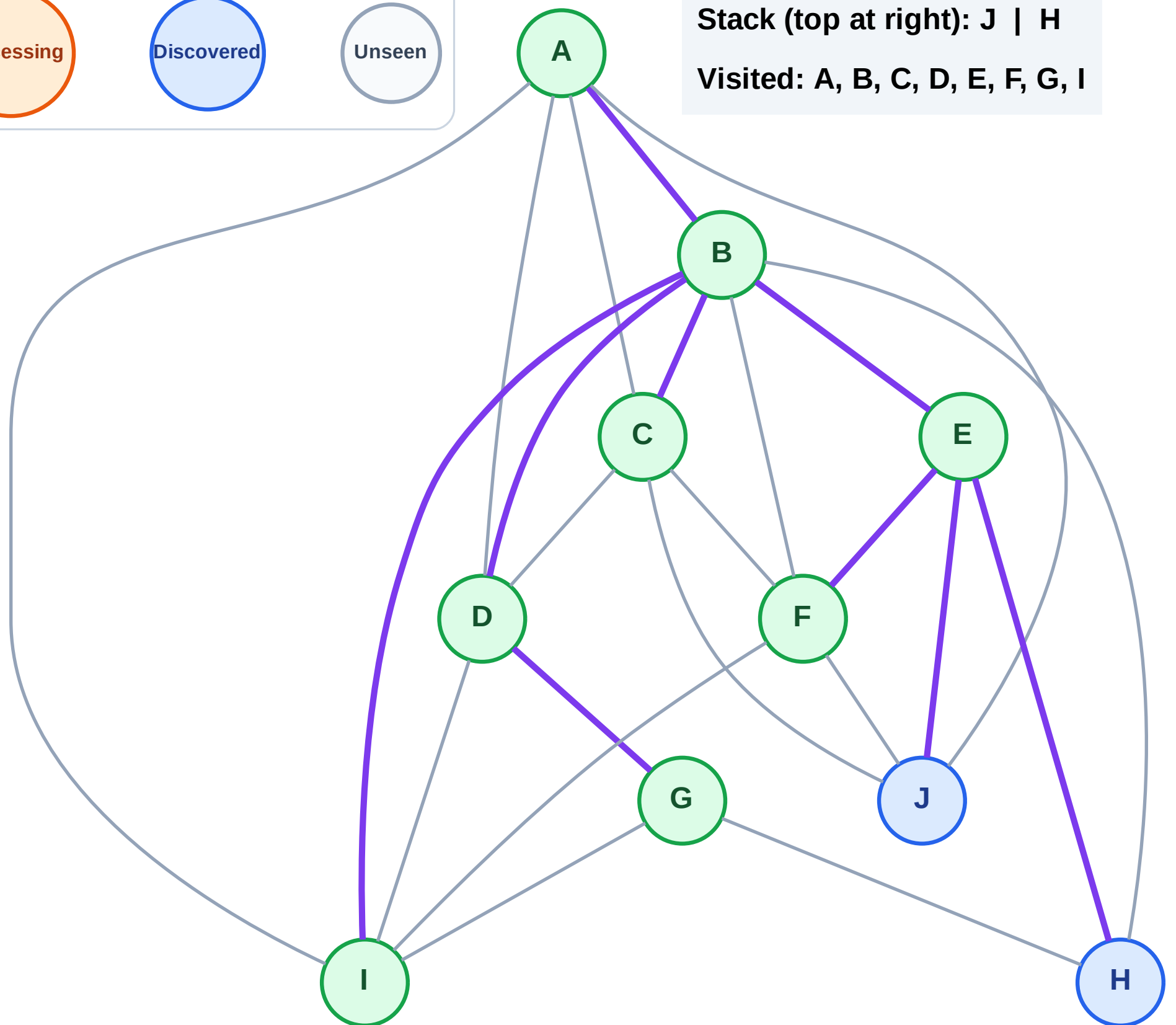
Step 17: No new neighbors from F

Legend



Stack (top at right): J | H

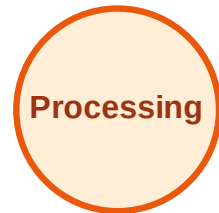
Visited: A, B, C, D, E, F, G, I



DFS Traversal

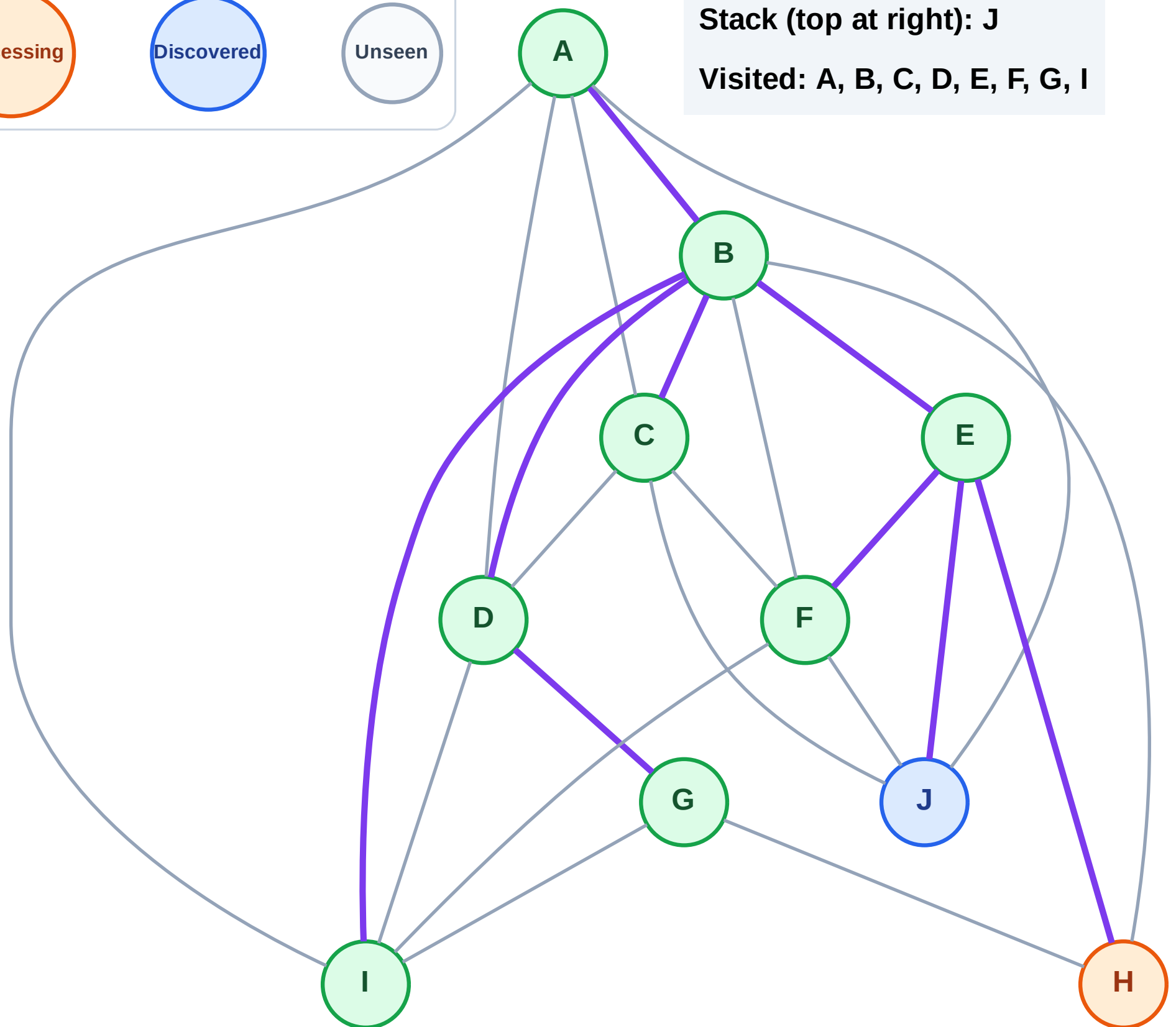
Step 18: Process node H

Legend



Stack (top at right): J

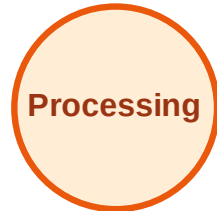
Visited: A, B, C, D, E, F, G, I



DFS Traversal

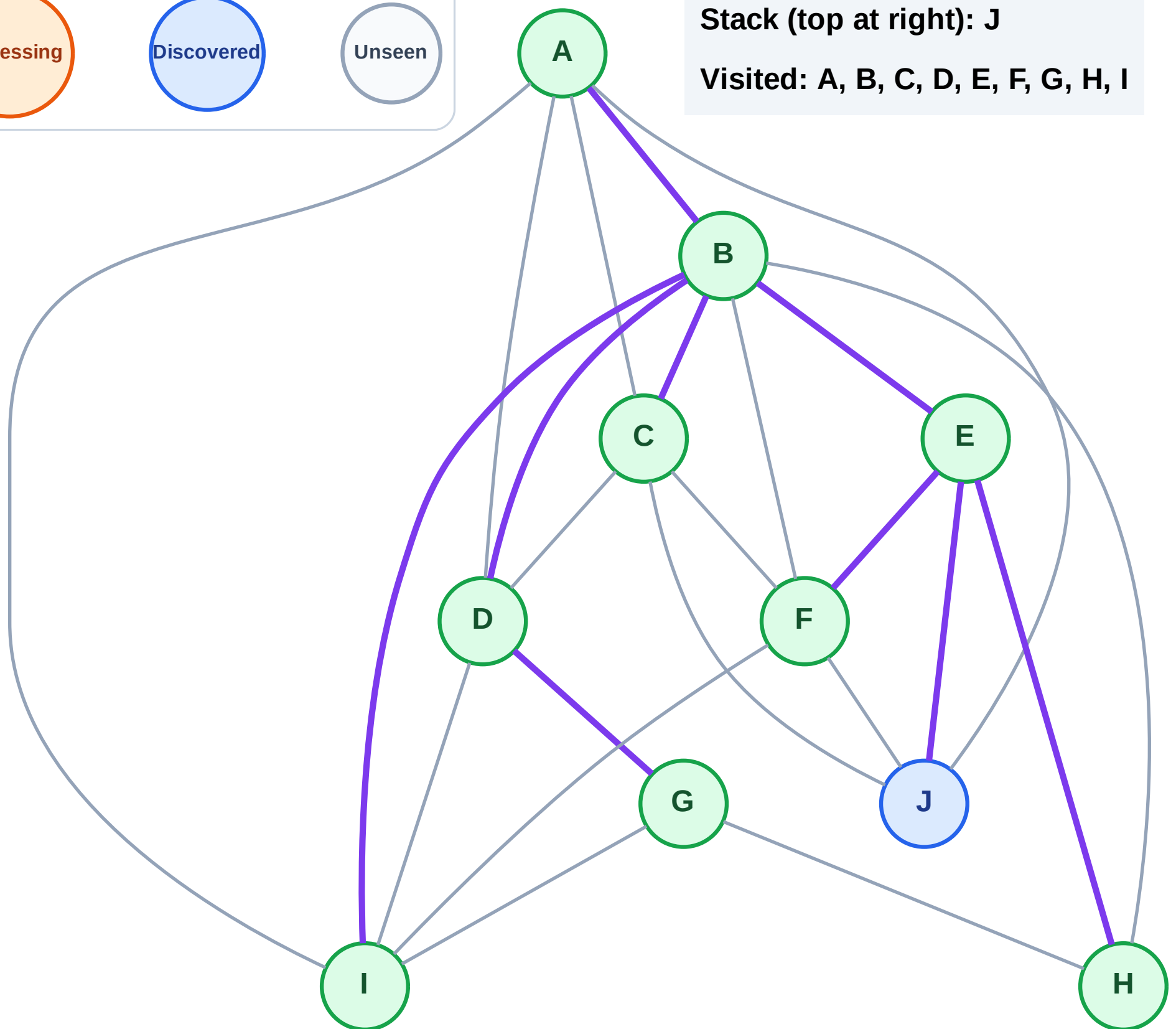
Step 19: No new neighbors from H

Legend



Stack (top at right): J

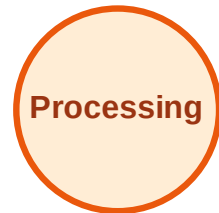
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

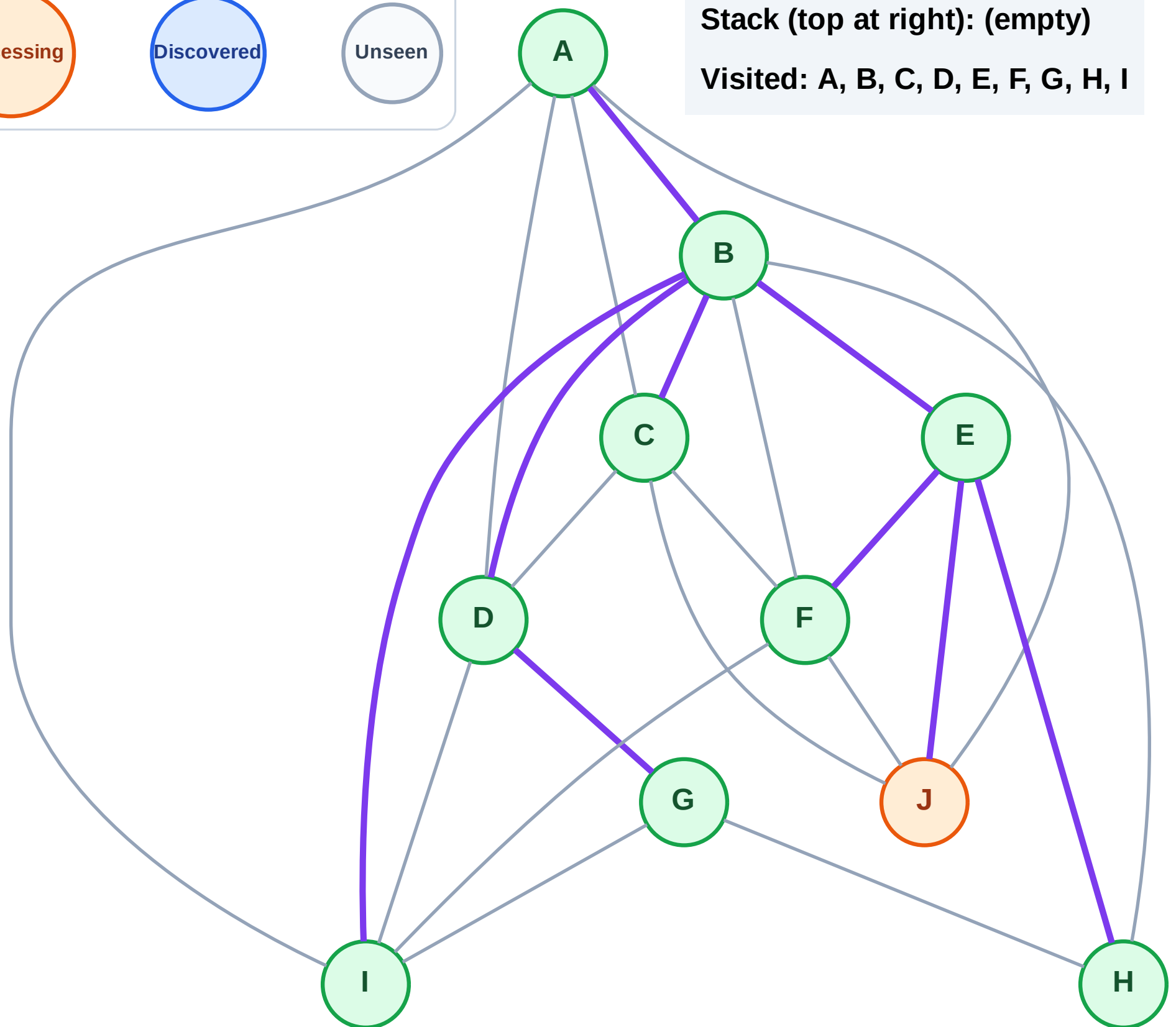
Step 20: Process node J

Legend



Stack (top at right): (empty)

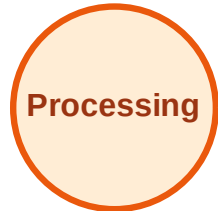
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

